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LINUX FUNDAMENTAL ON HACK THE BOX

1.1ntroduction

I have been learning Linux Fundamentals to build a strong foundation in the Linux operating system, widely used in servers, cloud environments, and security operations. I have gained skills in basic shell commands, scripting, and system monitoring, which help me manage systems, enumerate services, and respond to security incidents effectively.

1.2 Linux Structure

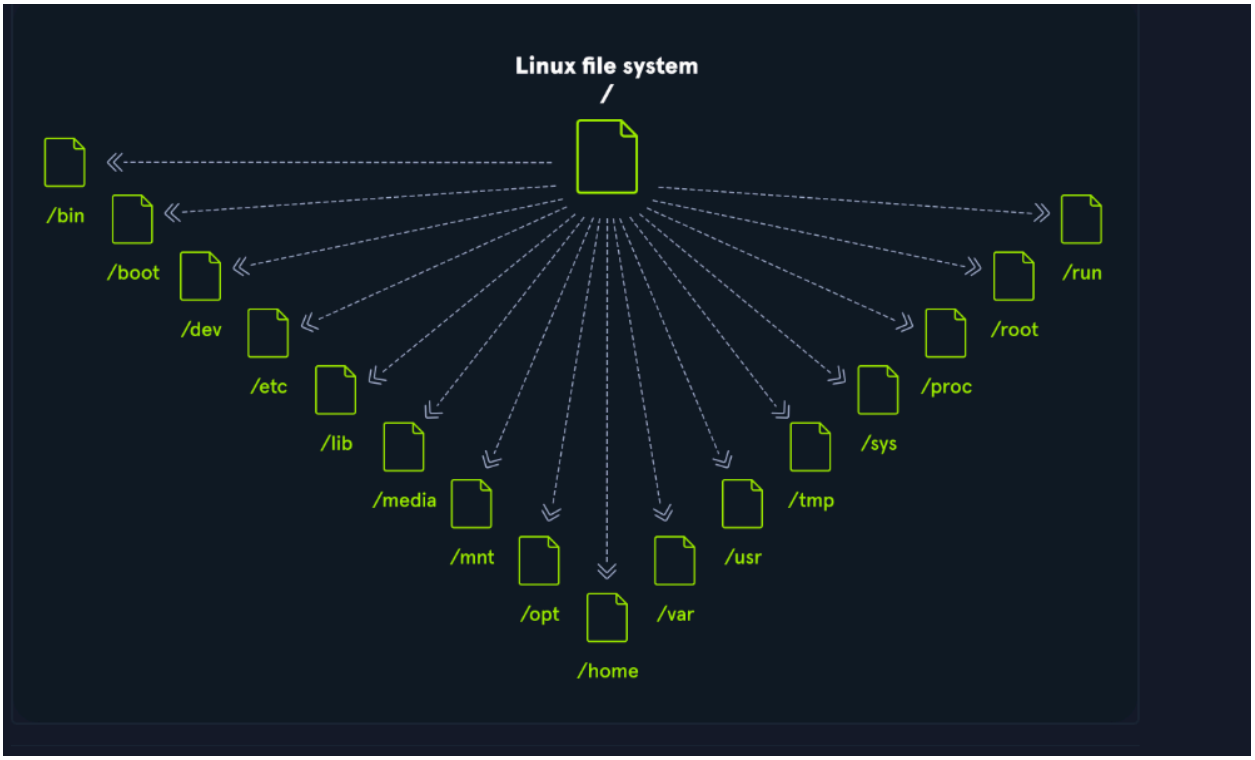
Linux is an open source operating system that manages all the hardware resources of a computer. Linux comes in many distributions known as Distros.Some of the most popular and well-known being Ubuntu, Debian, Fedora, OpenSUSE, elementary, Manjaro, Gentoo Linux, RedHat, and Linux Mint.there are different components of linux included in the diagram below:

Components

Component	Description
`Bootloader`	A piece of code that runs to guide the booting process to start the operating system. Parrot Linux uses the GRUB Bootloader.
`OS Kernel`	The kernel is the main component of an operating system. It manages the resources for system's I/O devices at the hardware level.
`Daemons`	Background services are called "daemons" in Linux. Their purpose is to ensure that key functions such as scheduling, printing, and multimedia are working correctly. These small programs load after we booted or log into the computer.
`OS Shell`	The operating system shell or the command language interpreter (also known as the command line) is the interface between the OS and the user. This interface allows the user to tell the OS what to do. The most commonly used shells are Bash, Tcsh/Csh, Ksh, Zsh, and Fish.
`Graphics server`	This provides a graphical sub-system (server) called "X" or "X-server" that allows graphical programs to run locally or remotely on the X-windowing system.
`Window Manager`	Also known as a graphical user interface (GUI). There are many options, including GNOME, KDE, MATE, Unity, and Cinnamon. A desktop environment usually has several applications, including file and web browsers. These allow the user to access and manage the essential and frequently accessed features and services of an operating system.
`Utilities`	Applications or utilities are programs that perform particular functions for the user or another program.

1.2.1 Linux File system

Linus has a tree like file system as illustrated in the diagram below:



1.2.2 Linux Distribution

Linux distributions, or distros, are operating systems built on the Linux kernel, each tailored for specific purposes such as servers, desktops, embedded devices, or mobile computing.. Popular examples include **Ubuntu, Fedora, CentOS, Debian, and Red Hat Enterprise Linux**, with **Kali Linux** being widely used in cybersecurity for its security-focused tools.

1.3 Introduction to Shell

A Linux terminal, also called a shell or command line, provides a text-based input/output (I/O) interface between users and the kernel for a computer system. The most commonly used shell in Linux is the Bourne-Again Shell (BASH), and is part of the GNU project.

1.4 Prompt Description

They simply show information like your username, the computer name and the directories or folders you are currently using.Other special characters are as illustrated in the diagram below:

Special Character	Description
\d	Date (Mon Feb 6)
\D{%Y-%m-%d}	Date (YYYY-MM-DD)
\H	Full hostname
\j	Number of jobs managed by the shell
\n	Newline
\r	Carriage return
\s	Name of the shell
\t	Current time 24-hour (HH:MM:SS)
\T	Current time 12-hour (HH:MM:SS)
\@	Current time
\u	Current username
\w	Full path of the current working directory

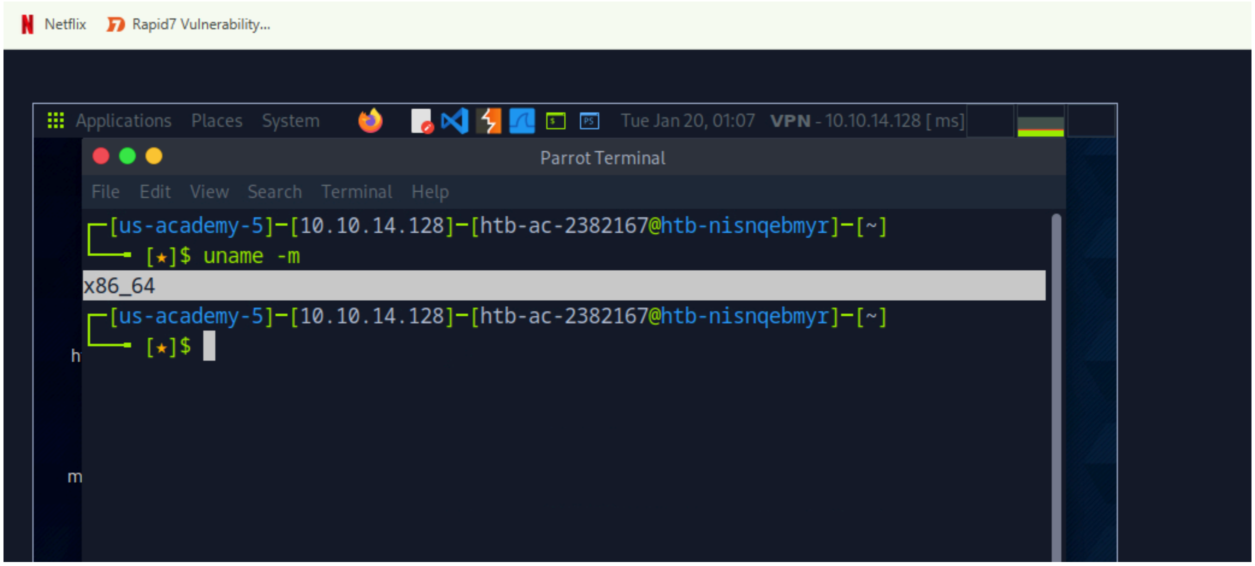
1.5 System Information

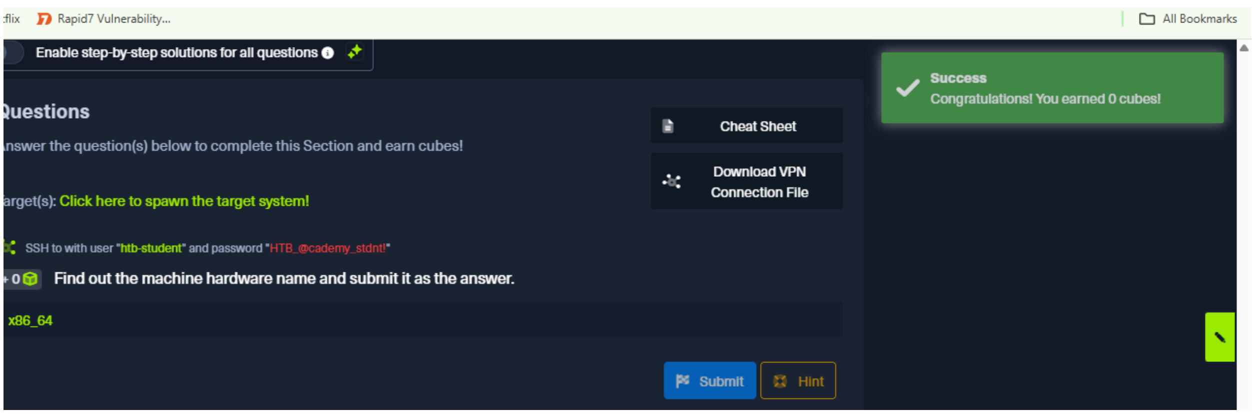
Question

1.Find out the machine hardware name and submit it as the answer.

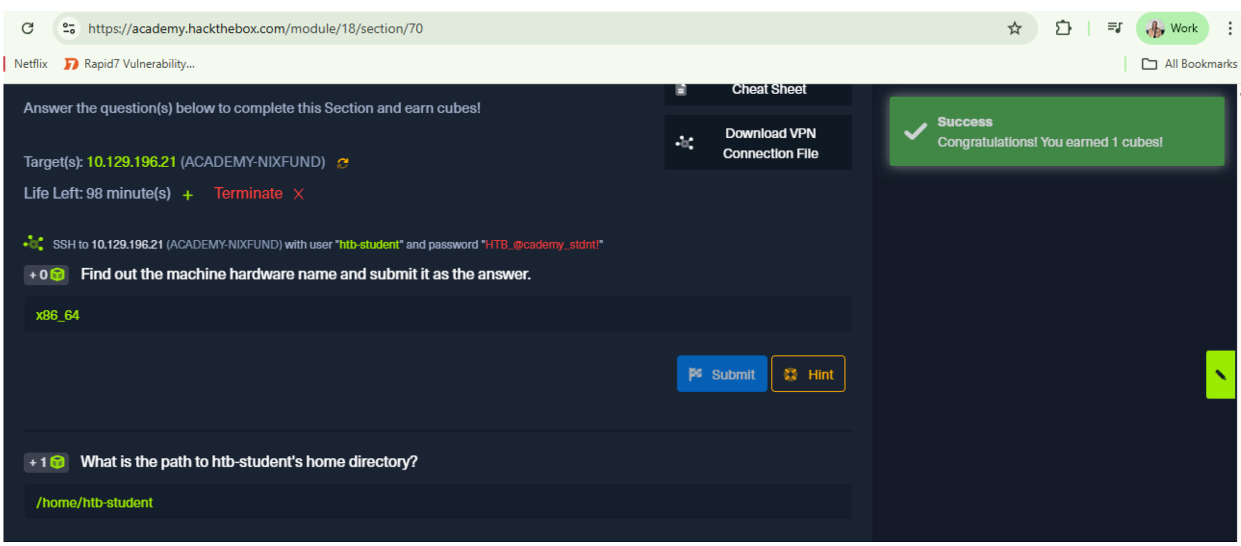
Answer:x86_64

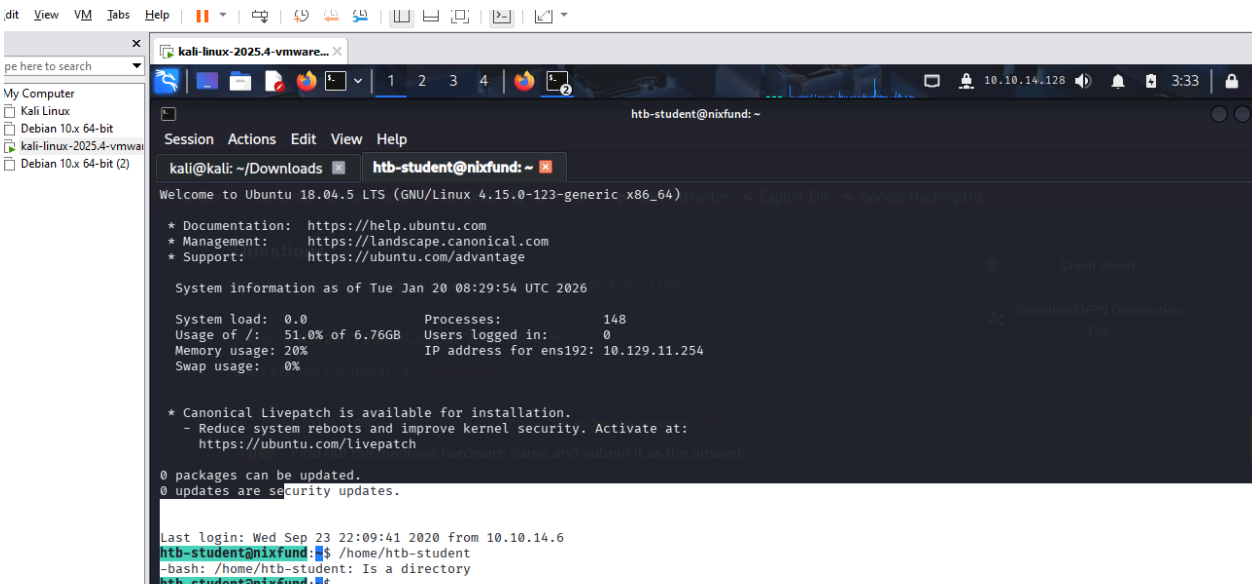
Explanation :I got the machine hardware name by running the command **"uname -m"**





2.What is the path to htb-student's home directory?
Answer:/home/htb-student

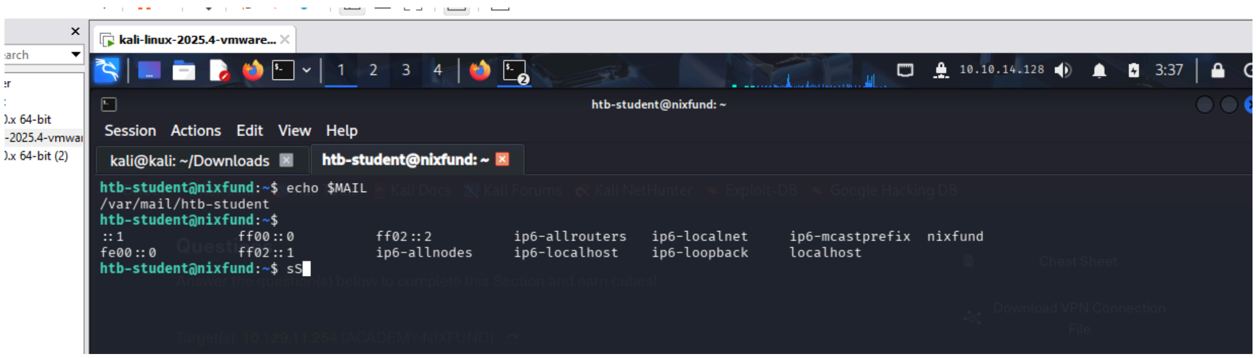


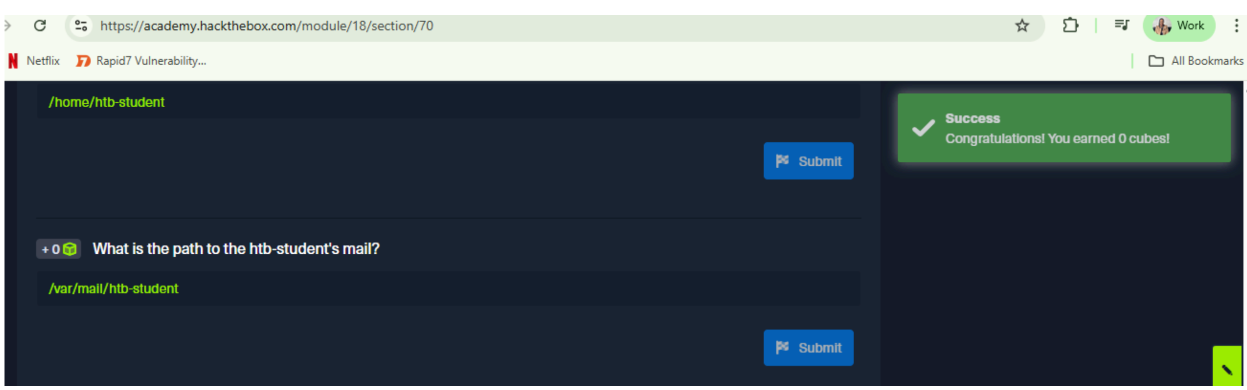


3.What is the path to the htb-student's mail?

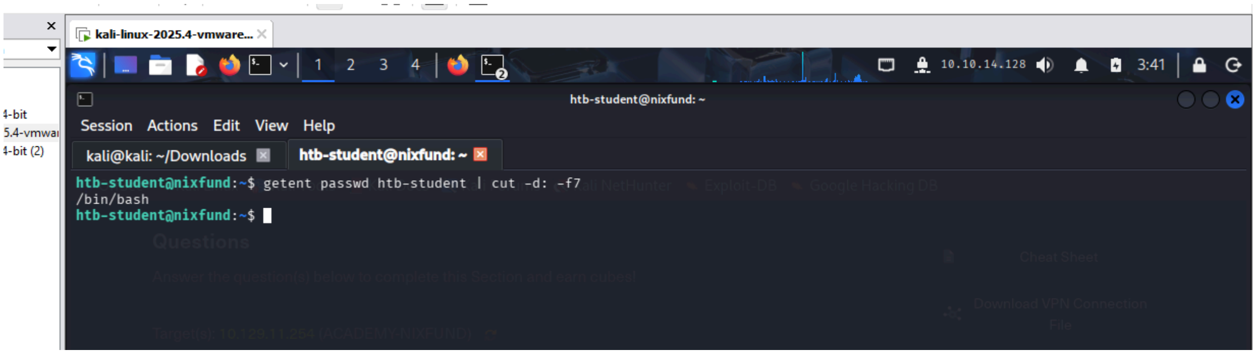
Answer:/var/mail/htb-student

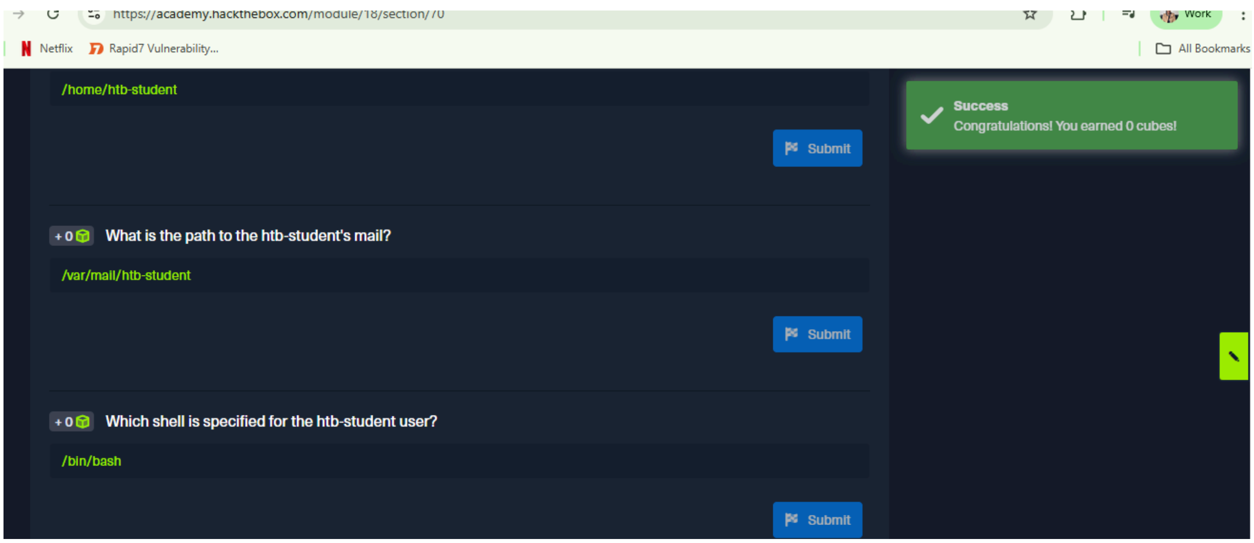
Explanation:After spawning the target , I ran the command echo $MAIL.



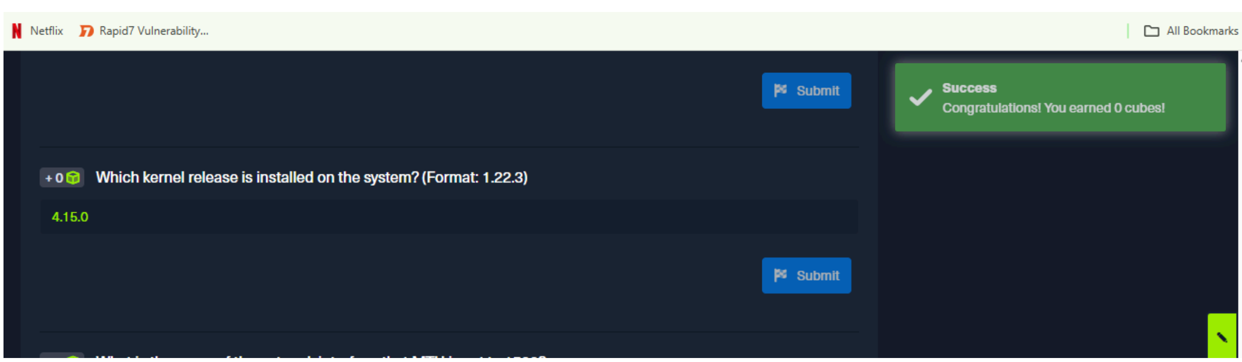


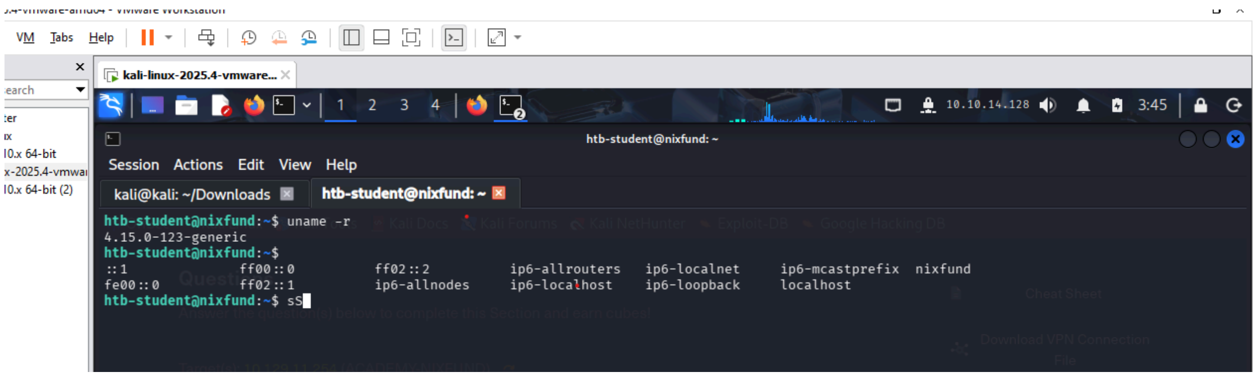
4.Which shell is specified for the htb-student user?
Answer:/bin/bash





5.Which kernel release is installed on the system? (Format: 1.22.3)
Answer:
4.15.0

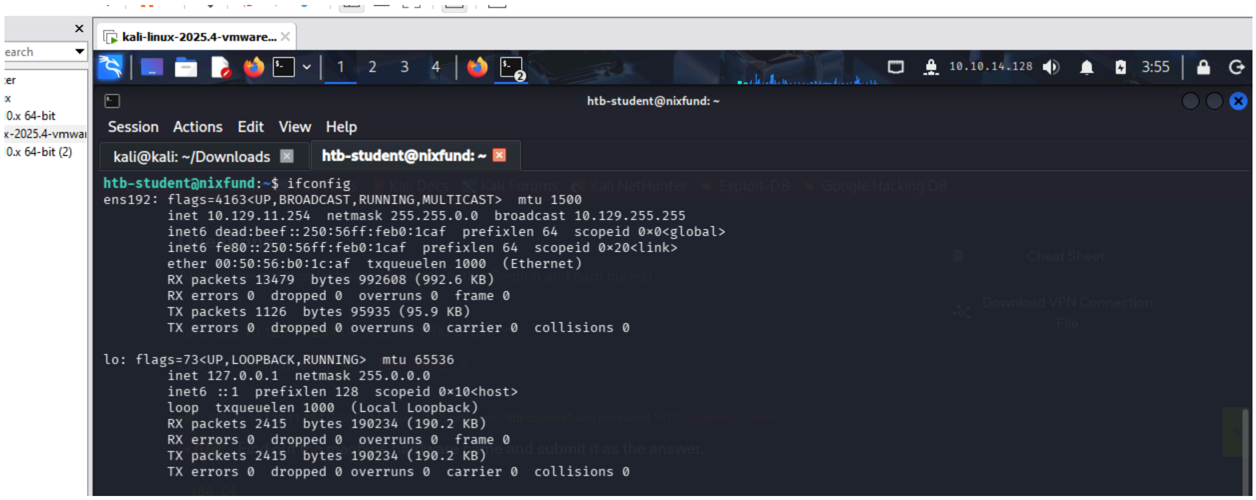


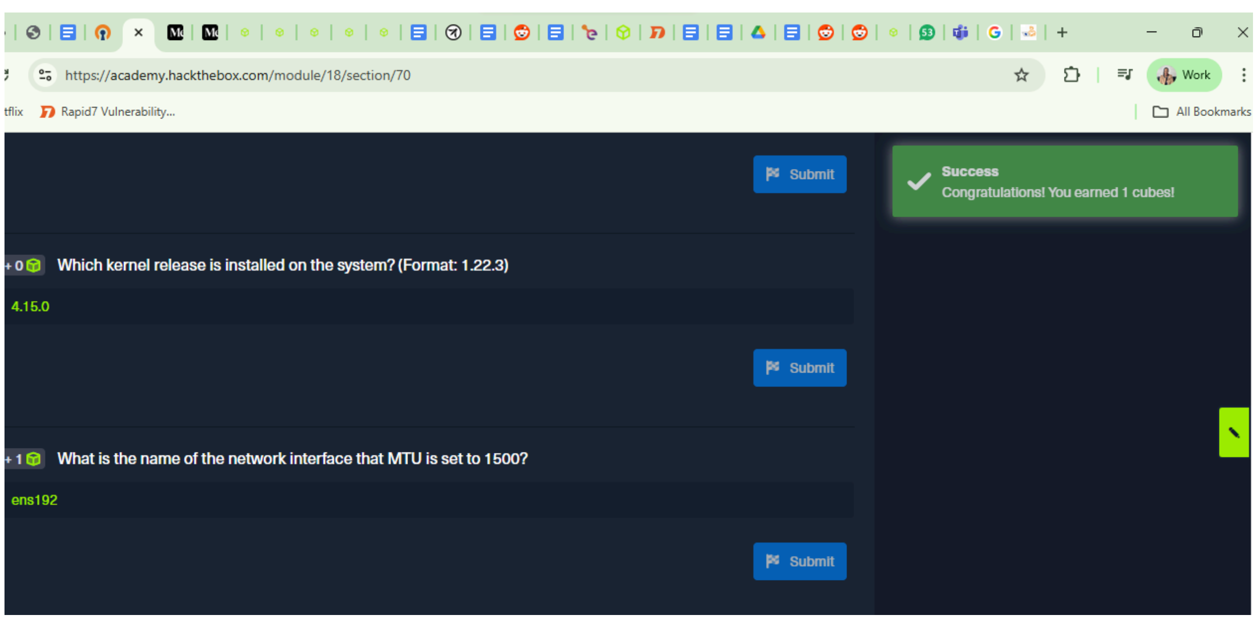


6.What is the name of the network interface that MTU is set to 1500?

Answer:ens192

Explanation ; i used the command ifconfig.





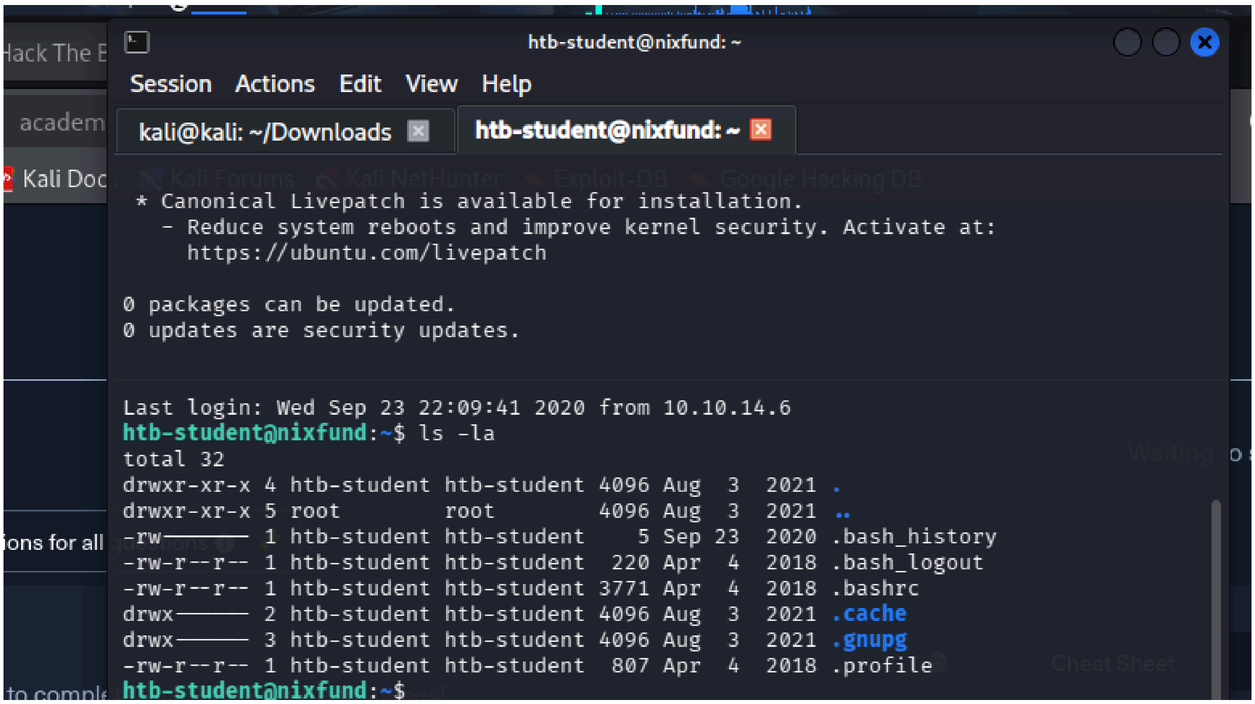
1.6 Navigation

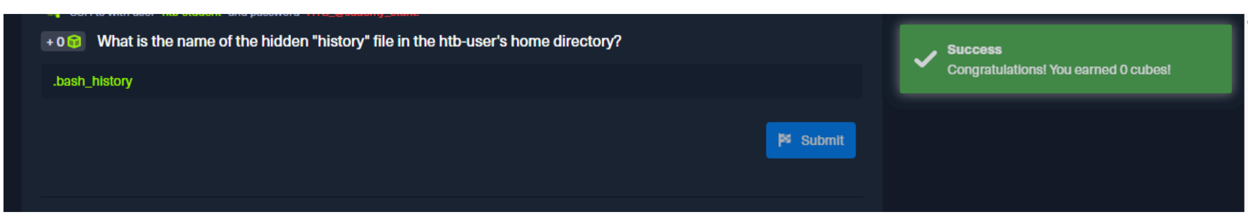
Basically how to navigate the linux console.use pwd, to know in which directory you are in. ls list all the directories and ls-la list all the hidden files.ls -l list more information on those directories.

Question

What is the name of the hidden "history" file in the htb-user's home directory?

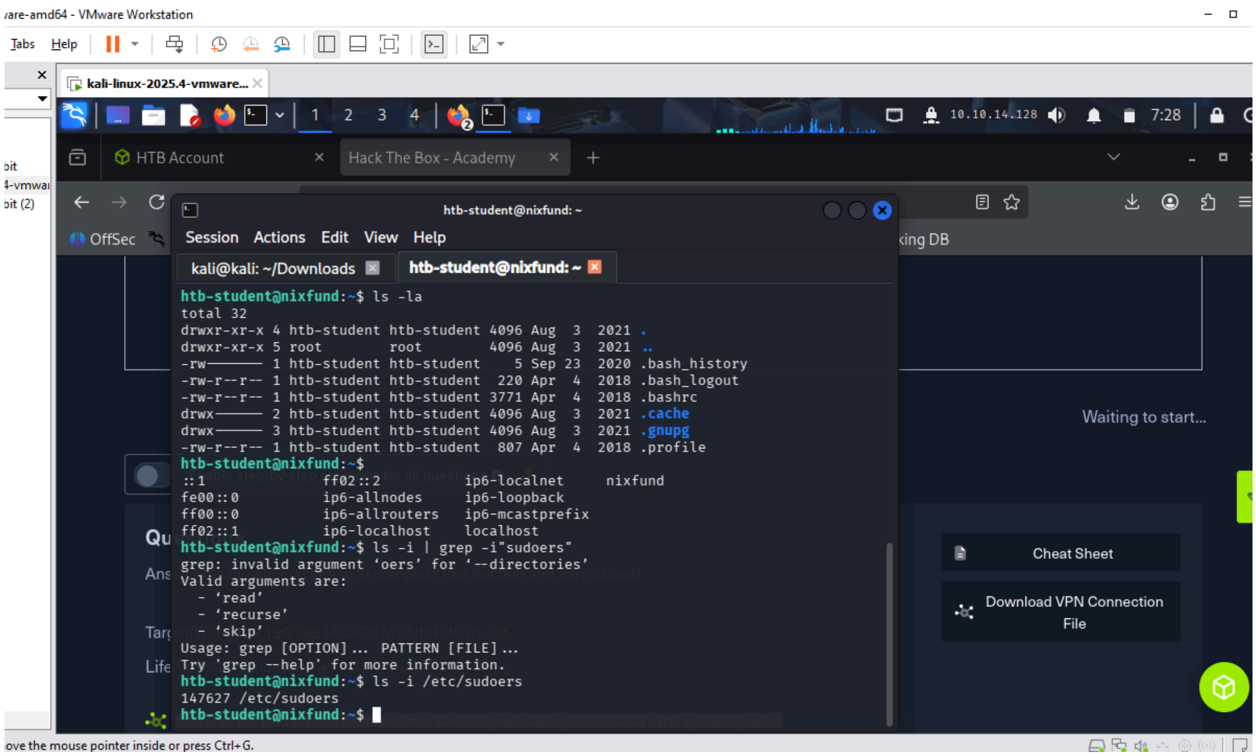
Answer: bash_history

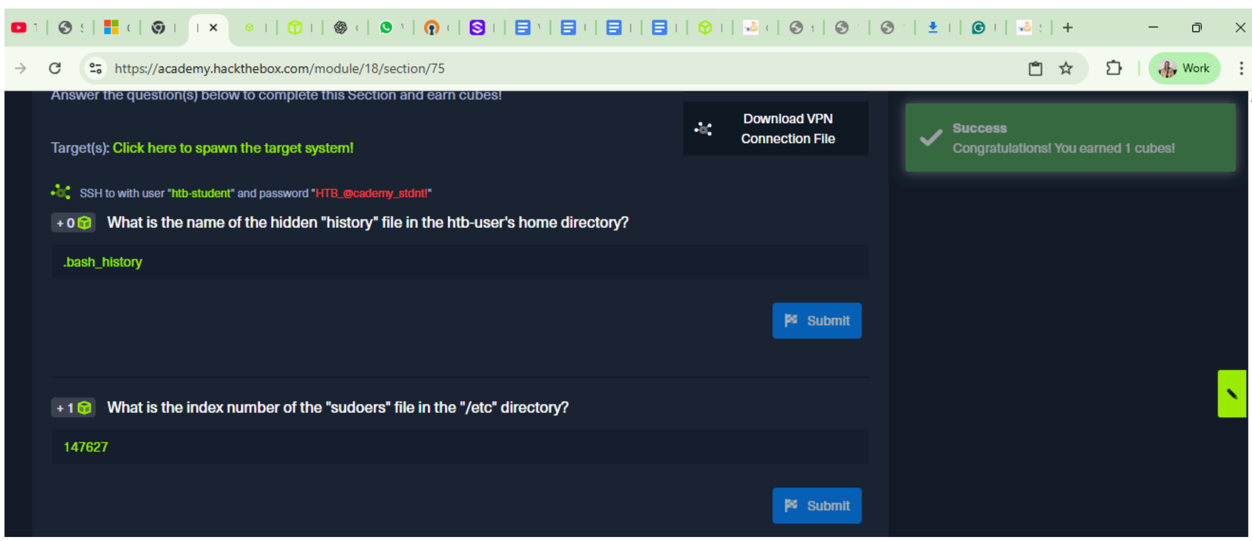




2.What is the index number of the "sudoers" file in the "/etc" directory?

Answer: 147627





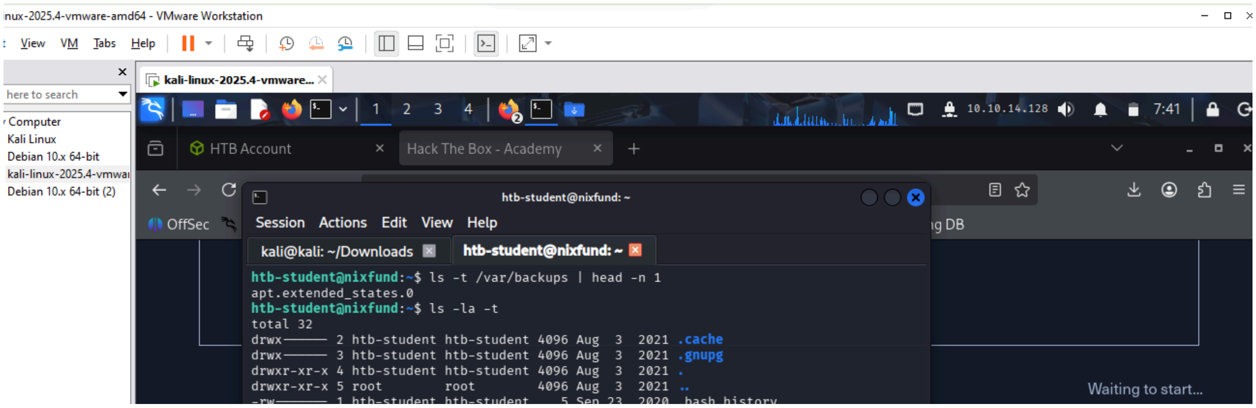
1.7 Working with files and Directories

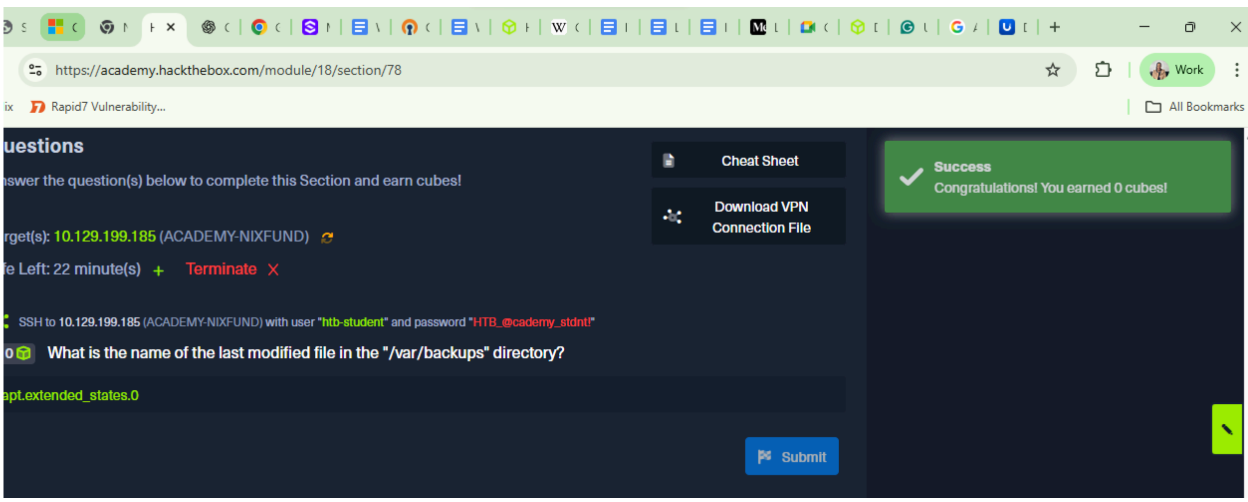
In Linux, files and directories are managed mainly through the terminal, which allows creating (touch), organizing (mkdir), moving (mv), and copying (cp) files efficiently. Commands like tree help visualize structures, and features like redirection and automation make batch operations faster. Mastering these tools streamlines system management and boosts efficiency compared to using graphical interfaces.

Question

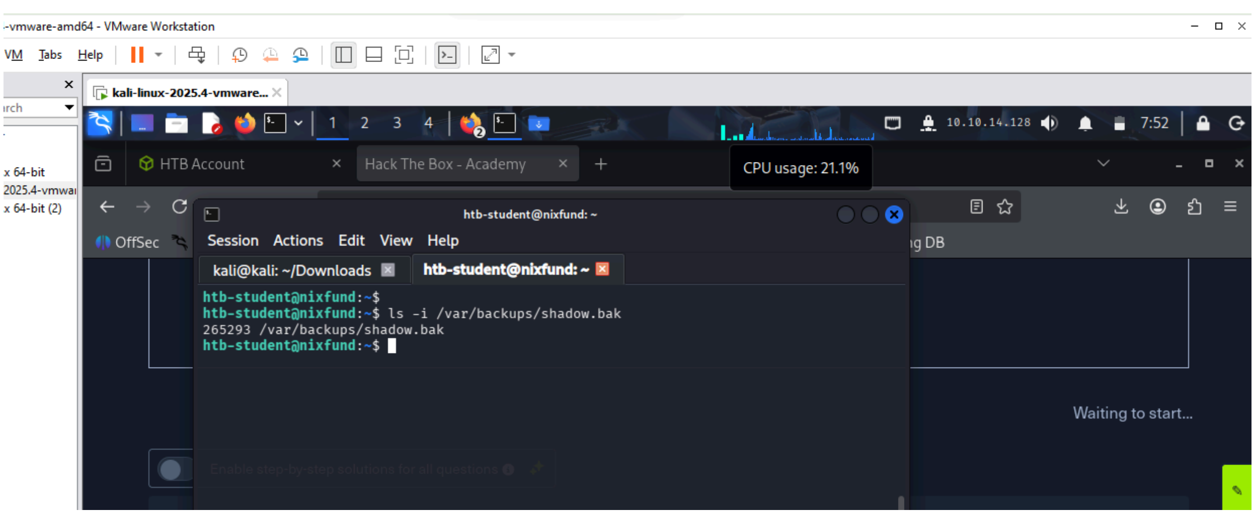
What is the name of the last modified file in the "/var/backups" directory?

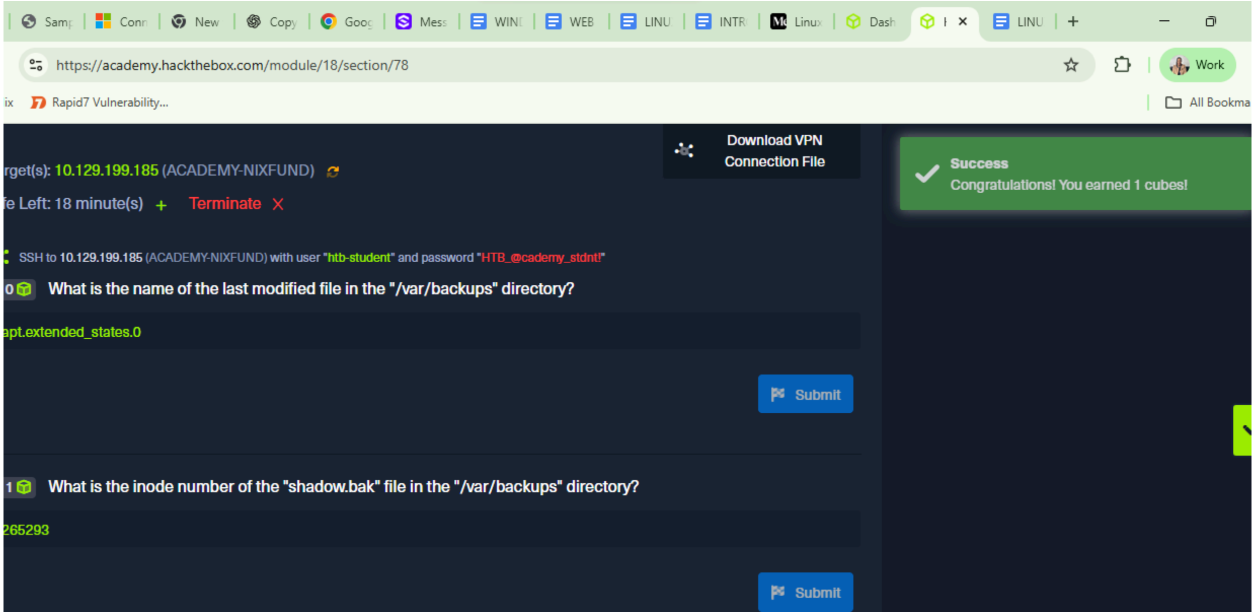
Answer: *apt.extended_states.0*





What is the inode number of the "shadow.bak" file in the "/var/backups" directory?
Answer: 265293





1.8 Editing Files

Linux files can be edited using Nano for simplicity or Vim for advanced control, and managing key system files like /etc/passwd and /etc/shadow is essential for security.

1.9 Find Files and Directories

Finding files and folders in Linux is essential for locating configuration files, scripts, and other important data. The command shows the path of a program, e.g.,which python/usr/bin/python.If the program doesn't exist, no result appears. Example:find / -type f -name *.conf -user root -size +20k -newermt 2020-03-03 -exec ls -al {} \; 2>/dev/null.For faster searches, locate uses a database updated with sudo updatedb, e.g., locate *.conf, but it offers fewer filters than find.

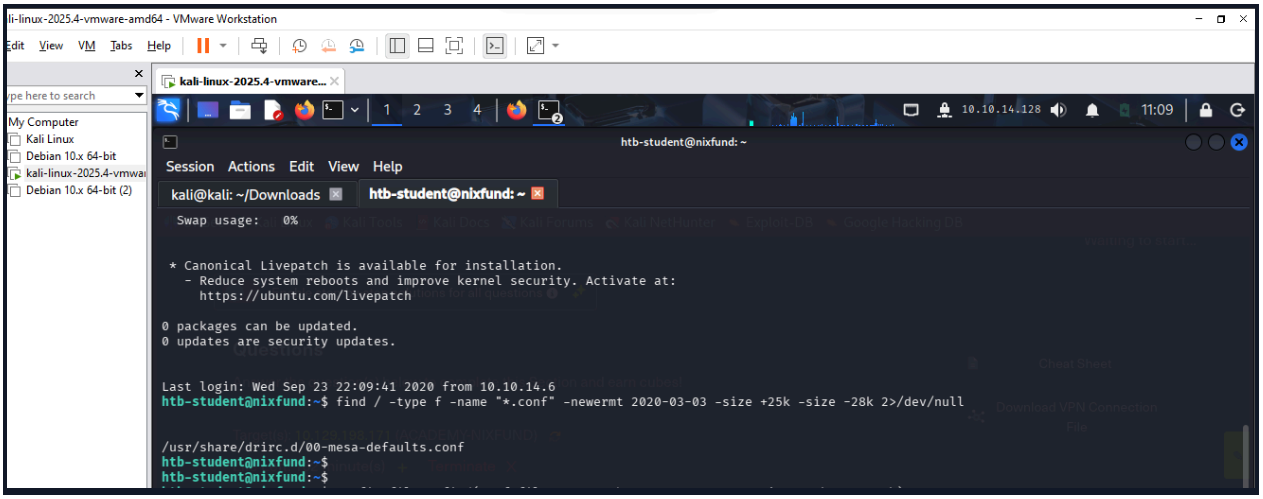
Question

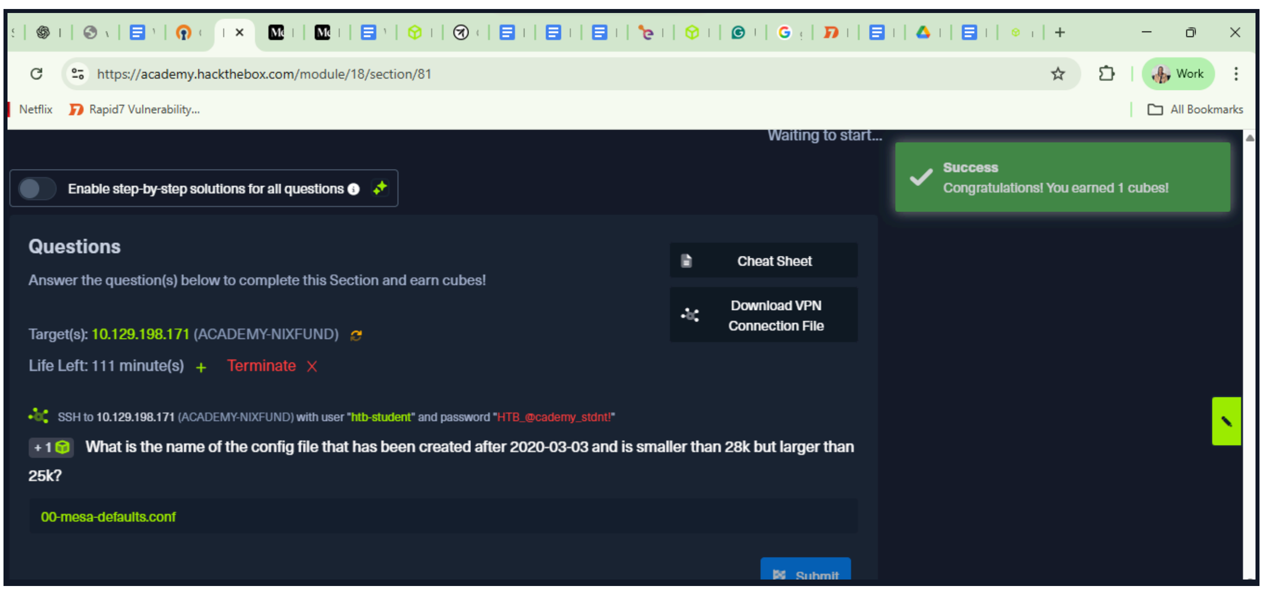
What is the name of the config file that has been created after 2020-03-03 and is smaller than 28k but larger than 25k?

Answer:

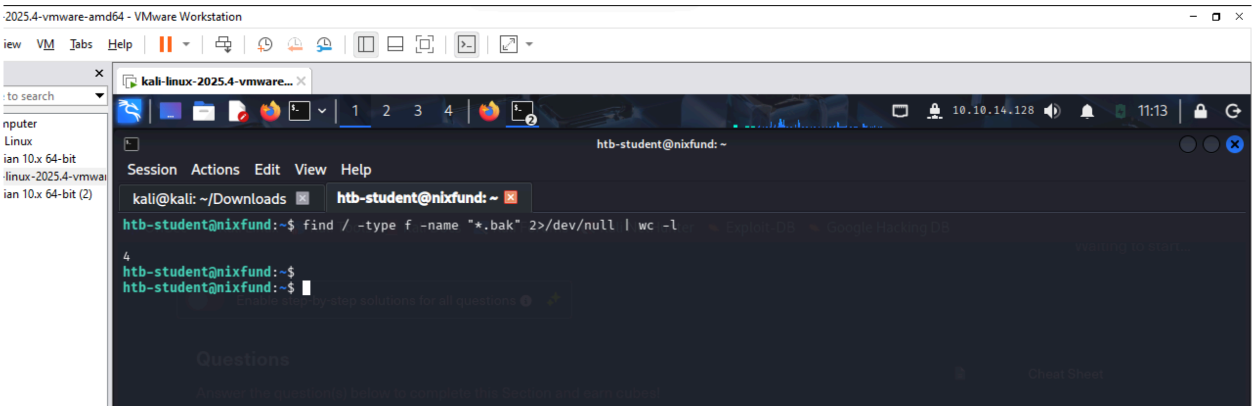
00-mesa-defaults.conf

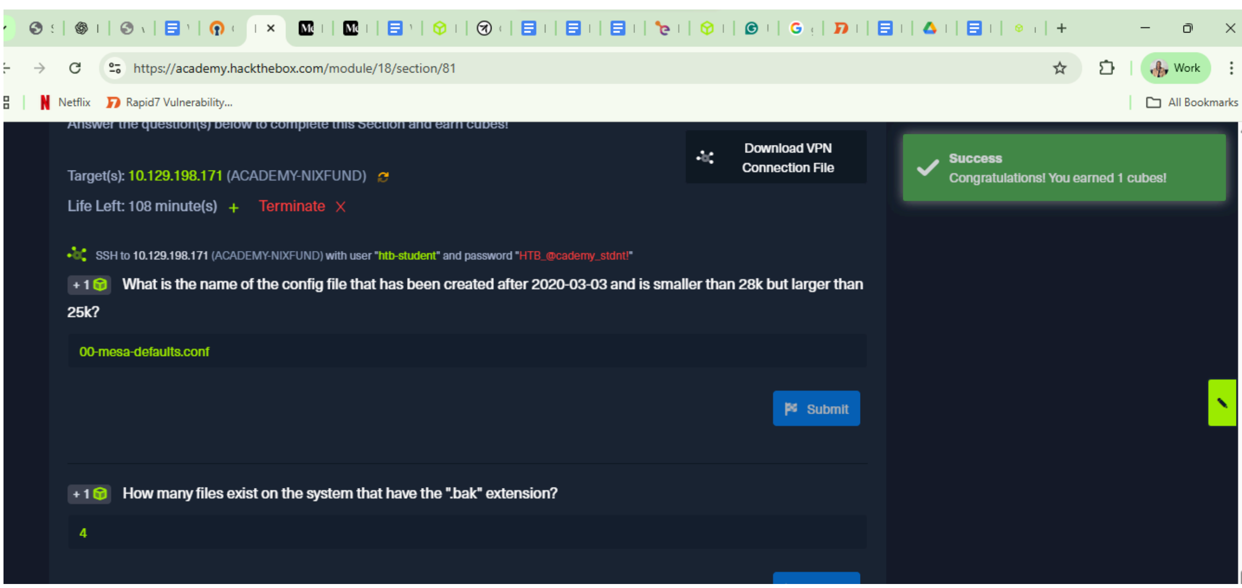
Explanation: I used this command to find the file including when it was created and what size.find /-type f -name "*.conf" -newermt 2020-03-03 -size +25k -size -28k 2>/dev/null


li-linux-2025.4-vmware-amd64 - VMware Workstation
htb-student@nixfund: ~
Session Actions Edit View Help
kali@kali: ~/Downloads
htb-student@nixfund: ~
Swap usage: 0%
* Canonical Livepatch is available for installation.
- Reduce system reboots and improve kernel security. Activate at:
https://ubuntu.com/livepatch
0 packages can be updated.
0 updates are security updates.
Last login: Wed Sep 23 22:09:41 2020 from 10.10.14.6
htb-student@nixfund:~$ find / -type f -name "*.conf" -newermt 2020-03-03 -size +25k -size -28k 2>/dev/null
/usr/share/drirc.d/00-mesa-defaults.conf
htb-student@nixfund:~$
htb-student@nixfund:~$


https://academy.hackthebox.com/module/18/section/81
Success
Congratulations! You earned 1 cubes!
Enable step-by-step solutions for all questions
Questions
Answer the question(s) below to complete this Section and earn cubes!
Cheat Sheet
Download VPN Connection File
Target(s): 10.129.198.171 (ACADEMY-NIXFUND)
Life Left: 111 minute(s) + Terminate X
SSH to 10.129.198.171 (ACADEMY-NIXFUND) with user "htb-student" and password "HTB_@cademy_stdnt!"
+1 What is the name of the config file that has been created after 2020-03-03 and is smaller than 28k but larger than 25k?
00-mesa-defaults.conf
Submit

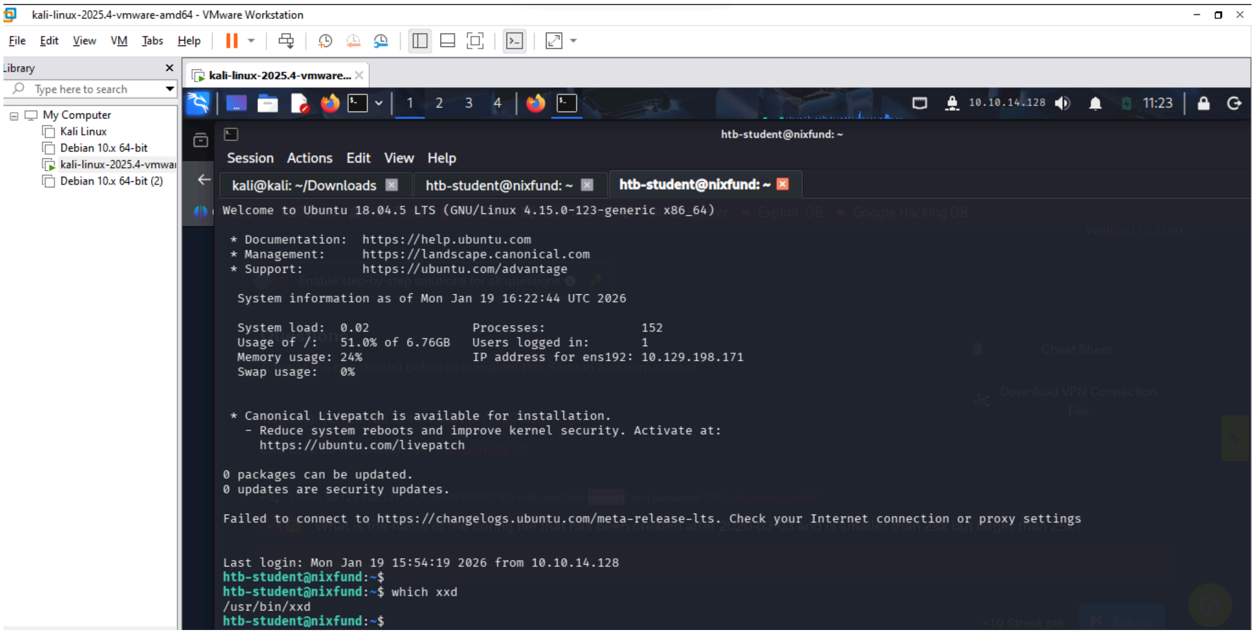
How many files exist on the system that have the ".bak" extension?
Answer:
4
Explanation:I ran this command on my target machine: find / type | name “*.bak” 1>/dev /null | wc 1

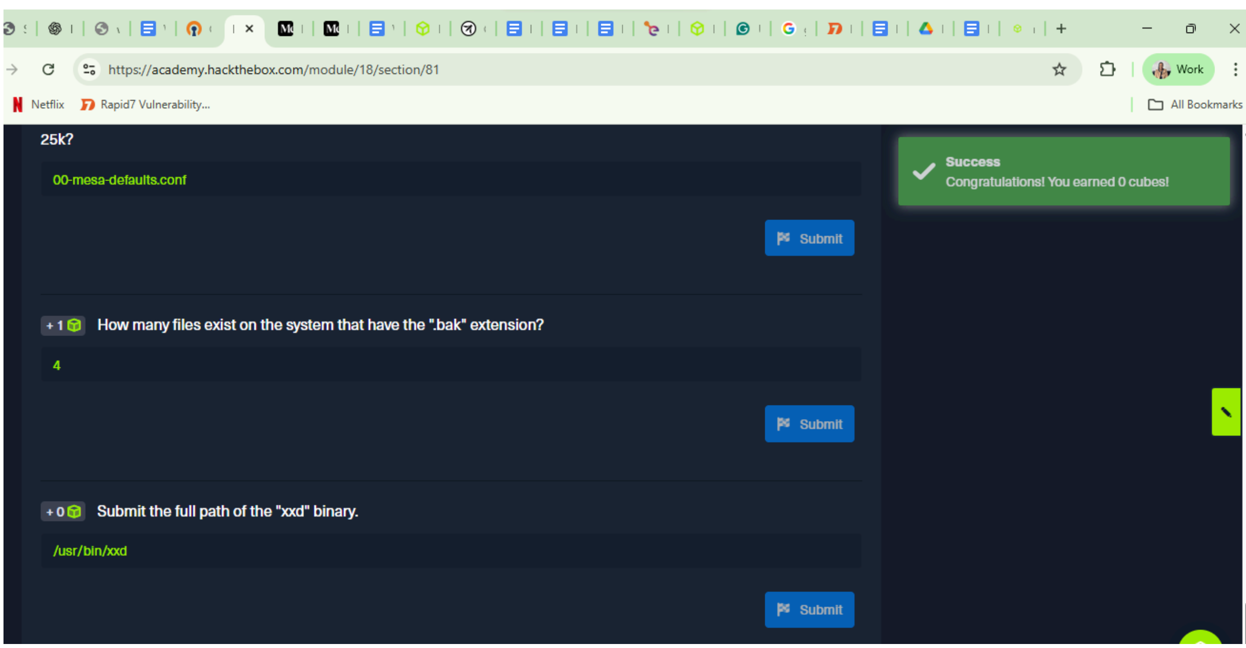




Submit the full path of the "xxd" binary.

Answer /usr/bin/xxd





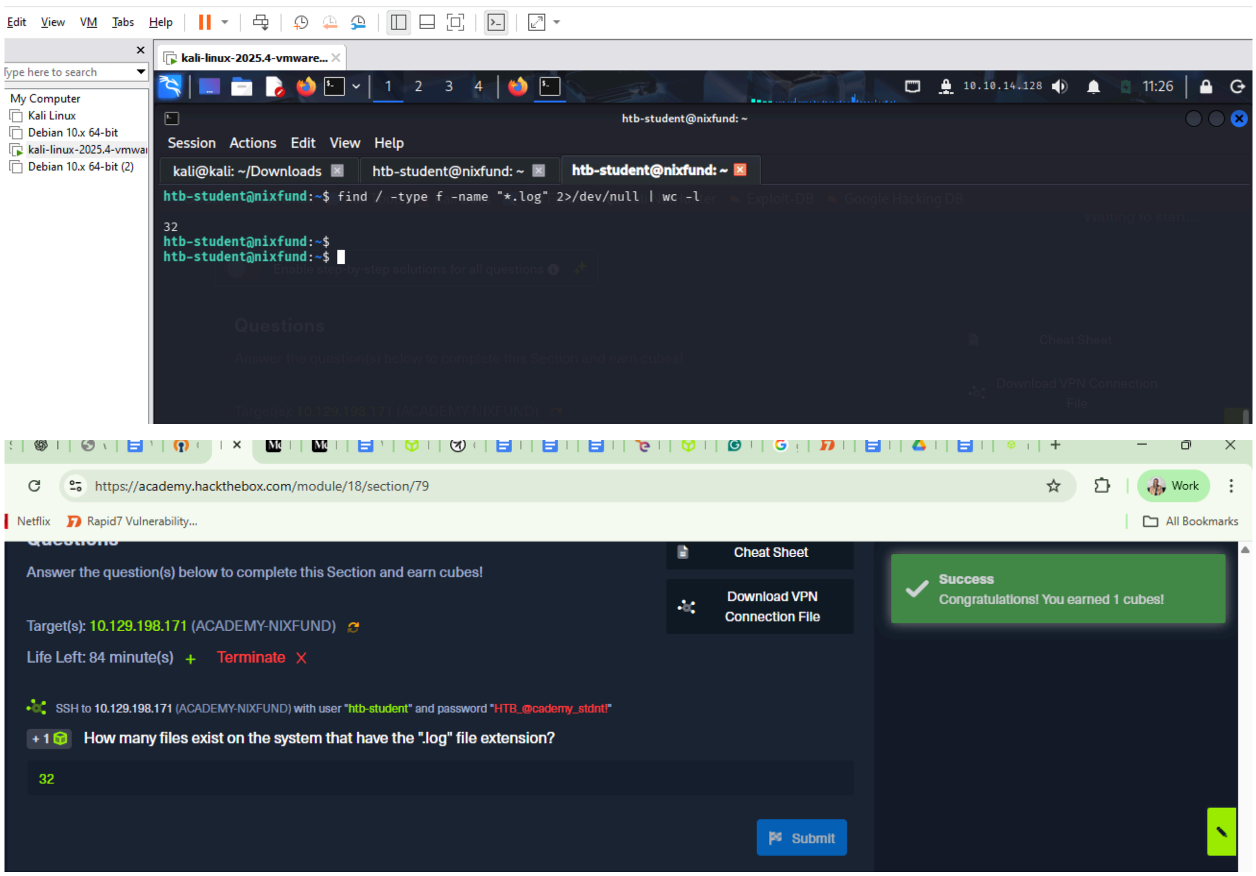
2.0 File descriptors and redirections

A file descriptor is a system reference for managing input or output in Linux: 0 = STDIN (input), 1 = STDOUT (output), 2 = STDERR (errors). You can redirect errors with 2>/dev/null, output to a file with > filename or append with >> filename, separate output and errors with 1> and 2>, read input from a file with < filename, or use a here-document with << EOF ... EOF. Pipes (|) send the STDOUT of one command to another, e.g., find /etc/ -name *.conf 2>/dev/null | grep systemd | wc -l.

Questions

1 How many files exist on the system that have the ".log" file extension?

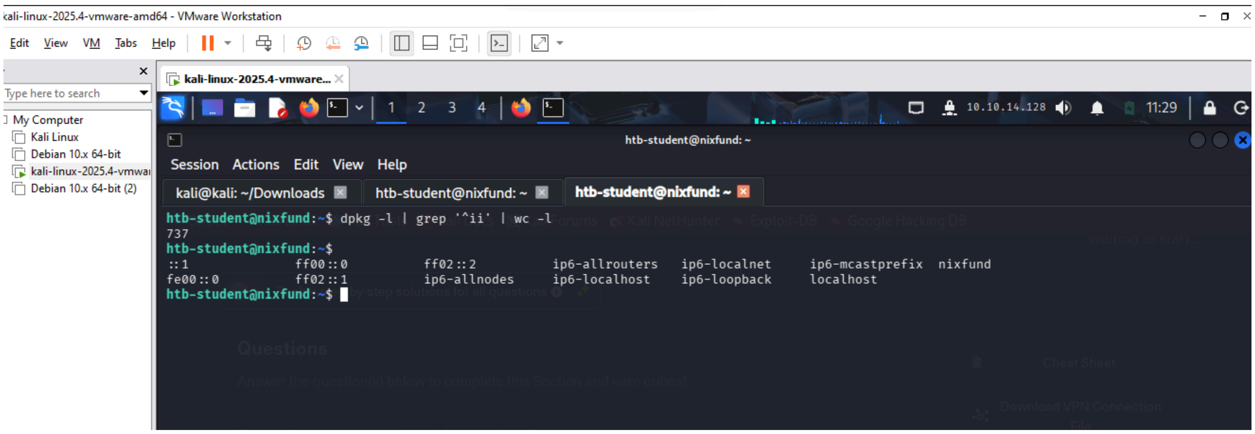
Answer :32

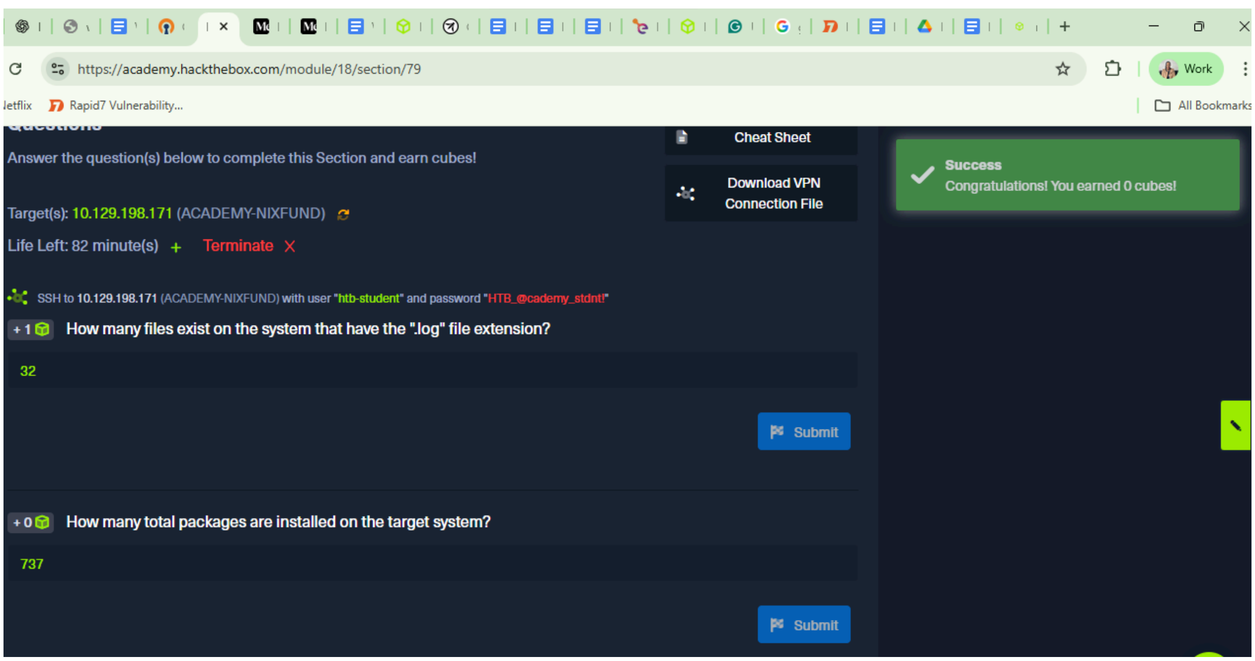


2. How many total packages are installed on the target system?

Answer:737

I ran the below command as shown in the screenshot.





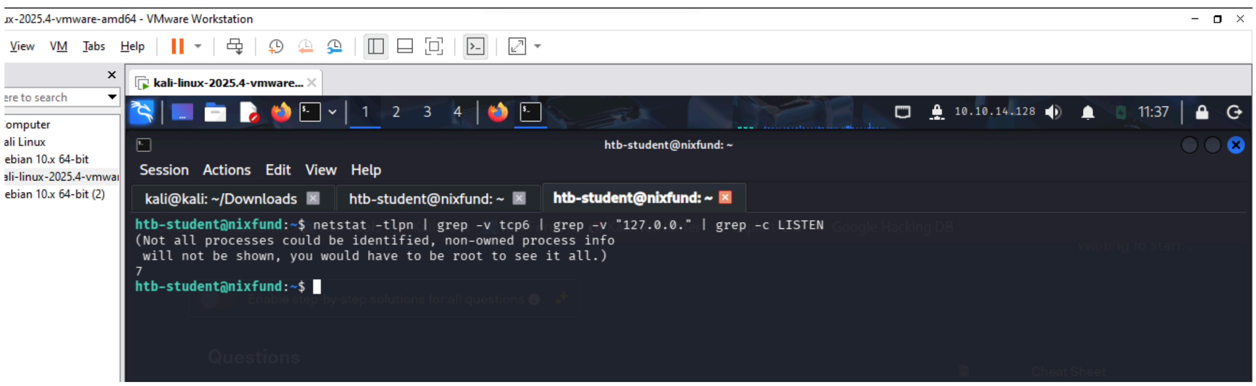
2.1 Filter Contents

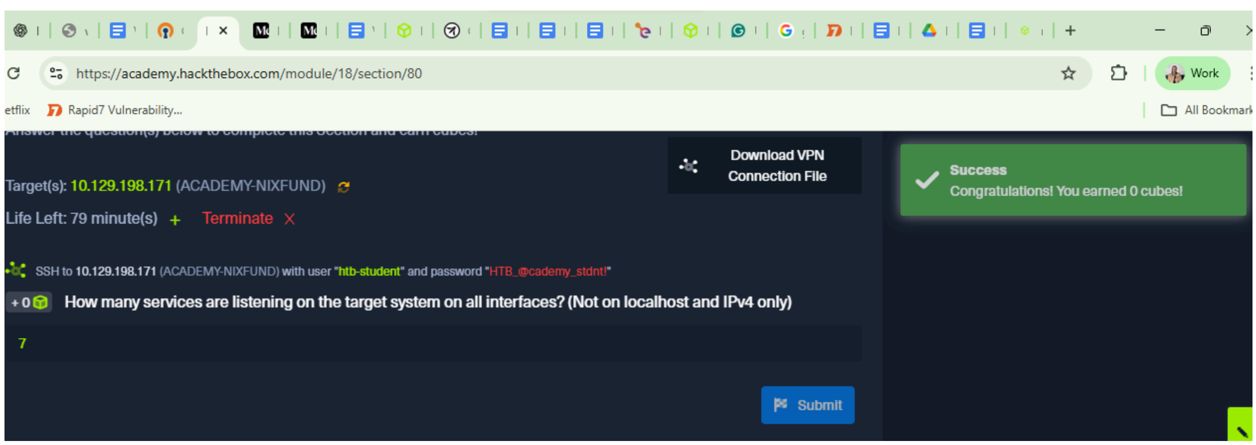
Linux tools like more/less view files, head/tail show start/end lines, sort orders output, grep filters patterns, cut and tr extract/replace fields, awk selects columns, sed substitutes text, and wc -l counts lines. Combined in pipelines, they let you efficiently filter and process /etc/passwd data.

Questions

1.How many services are listening on the target system on all interfaces? (Not on localhost and IPv4 only)

Answer :7

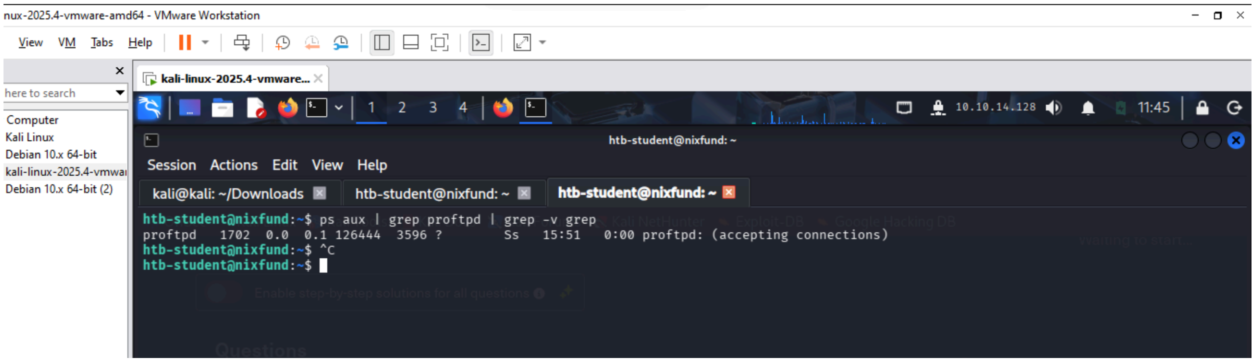


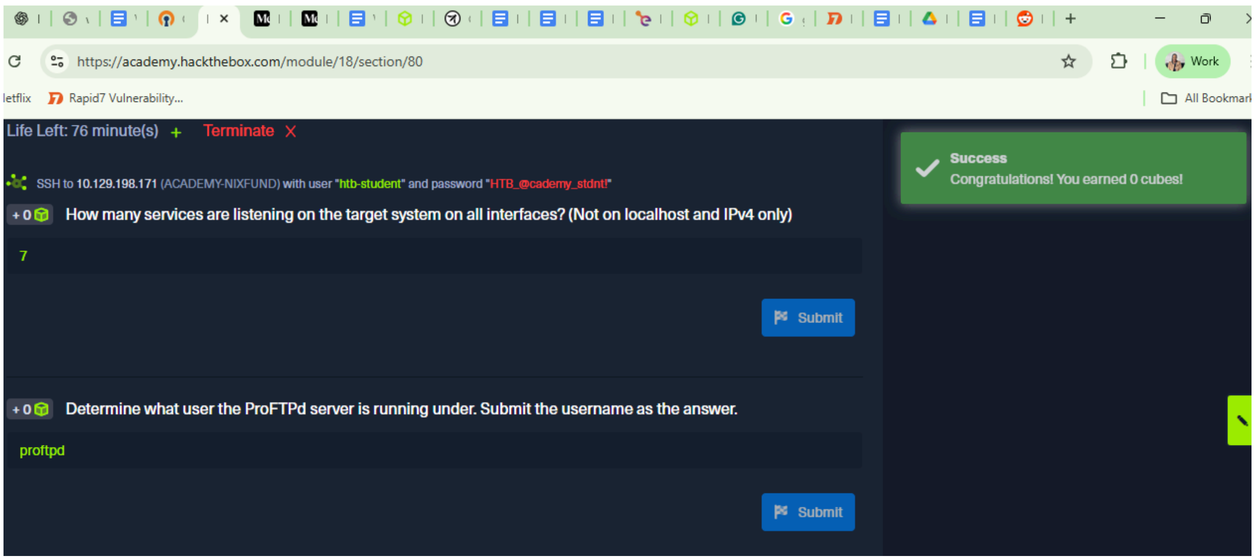


2.Determine what user the ProFTPd server is running under. Submit the username as the answer.

Answer

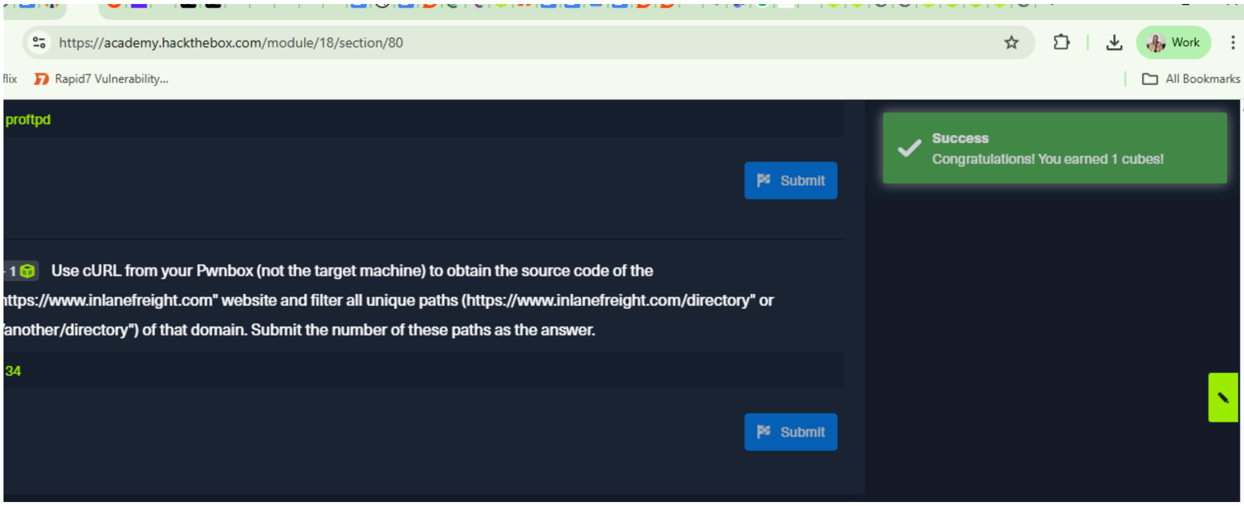
proftpd

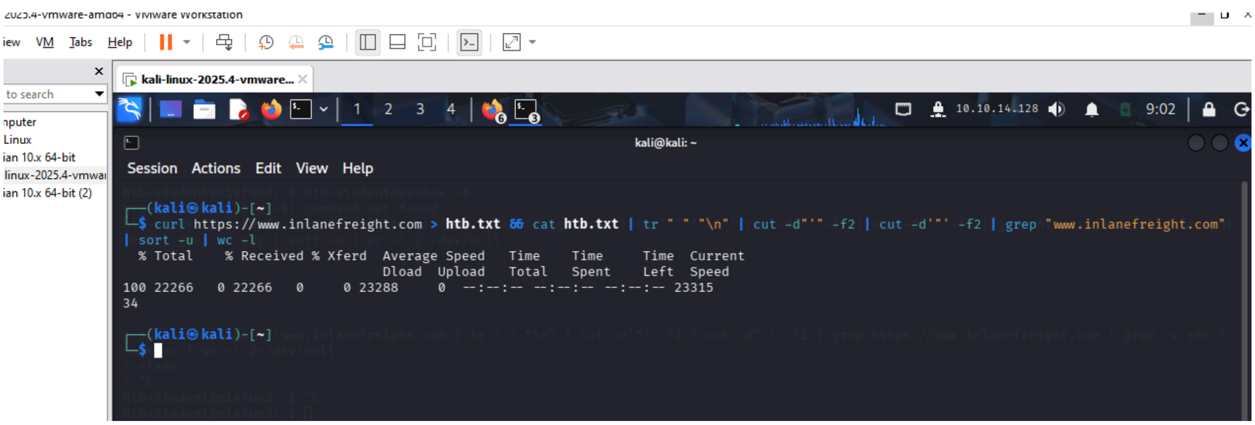




3.Use cURL from your Pwnbox (not the target machine) to obtain the source code of the "https://www.inlanefreight.com" website and filter all unique paths (https://www.inlanefreight.com/directory" or "/another/directory") of that domain. Submit the number of these paths as the answer.

Answer:34





2.2 Regular Expressions

Regular expressions (RegEx) are patterns for searching and manipulating text using symbols like ( ) for grouping, [ ] for character classes, { } for repetition, and | for OR. Tools like grep use RegEx to filter text, e.g., finding lines with one or multiple patterns, ignoring comments, or matching specific settings in files like sshd_config.

2.3 Permission Management

Linux permissions control who can read, write, or execute files and directories. chmod changes permissions, chown changes ownership, octal values represent permission sets, SUID/SGID allow running programs with owner/group rights, and the sticky bit restricts file deletion in shared directories.

2.4 User Management

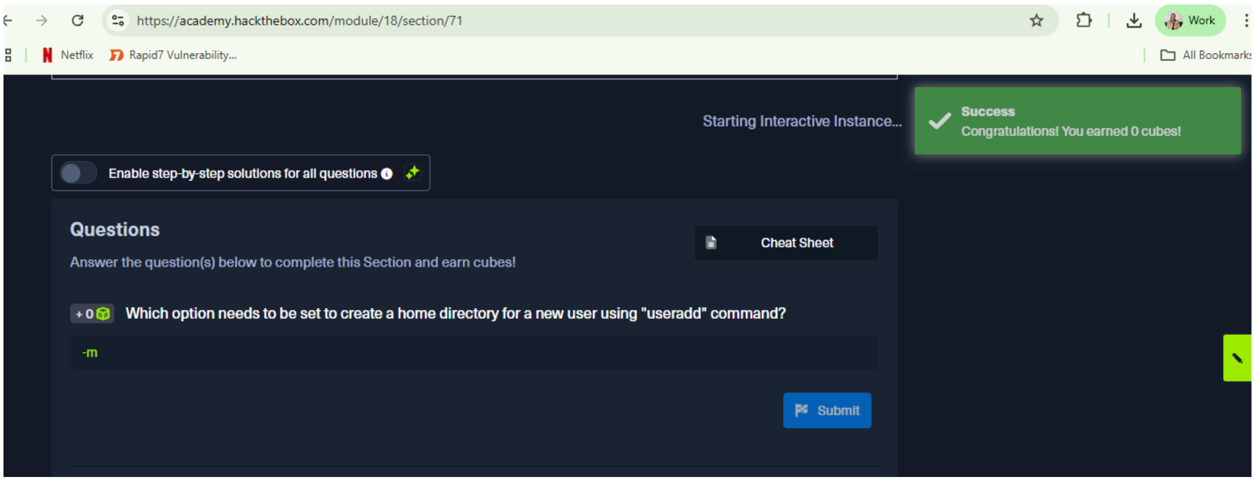
Linux user management lets you create users, assign groups, and run commands with proper privileges using sudo or su. Commands like useradd, usermod, userdel, and passwd control access, helping maintain system security.

Question

1.Which option needs to be set to create a home directory for a new user using "useradd" command?

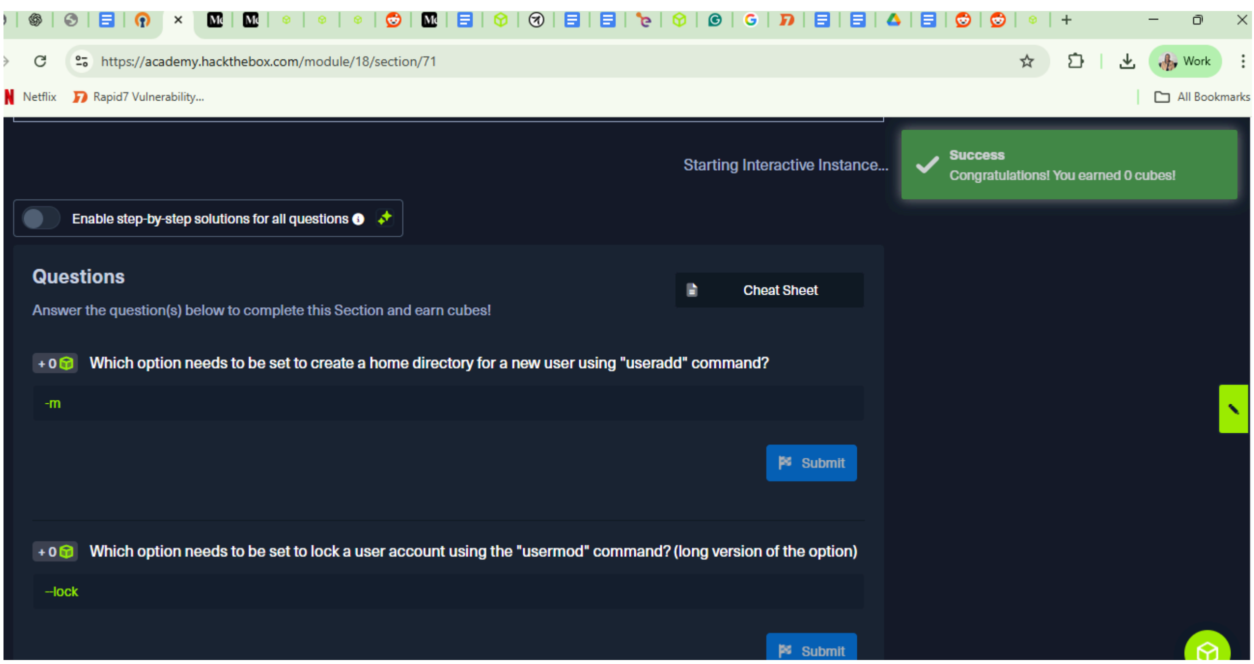
Answer:-m

Short for “make home”



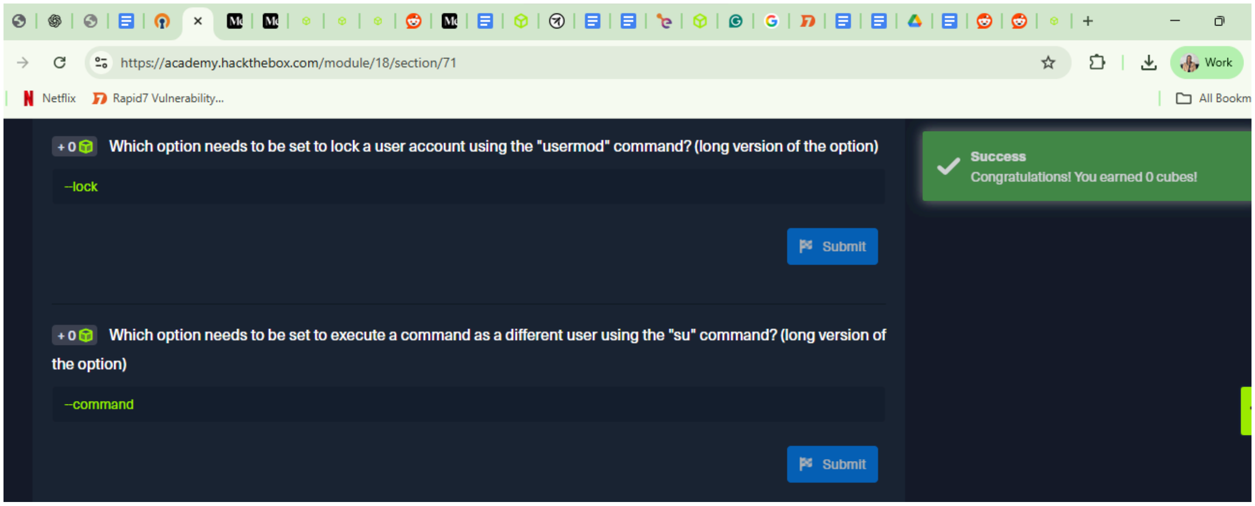
2.Which option needs to be set to lock a user account using the "usermod" command? (long version of the option)

Answer: –lock



3.Which option needs to be set to execute a command as a different user using the "su" command? (long version of the option)

Answer:--command



2.5 Package Management

Linux package managers let you install, update, and remove software while handling dependencies. Debian-based systems use **APT/dpkg**, Red Hat uses **yum/rpm**, and others include **snap, pip, gem, and git**. Commands like apt install, apt-cache search, and dpkg -i manage packages, while Git can clone projects from repositories.

2.6 Service and process management

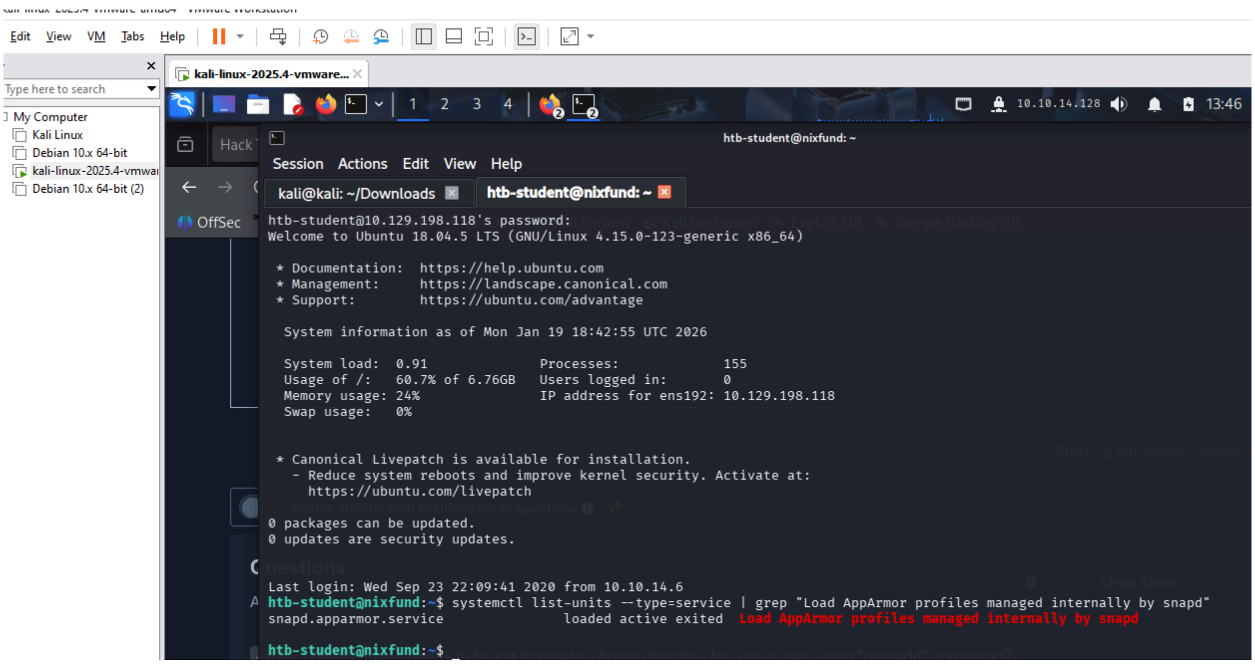
Linux services (daemons) run in the background to keep the system working, are managed with systemctl (start/stop/status/enable), monitored via journalctl and ps, controlled with signals using kill, and processes can be backgrounded or chained together using &, ;, &&, and |.

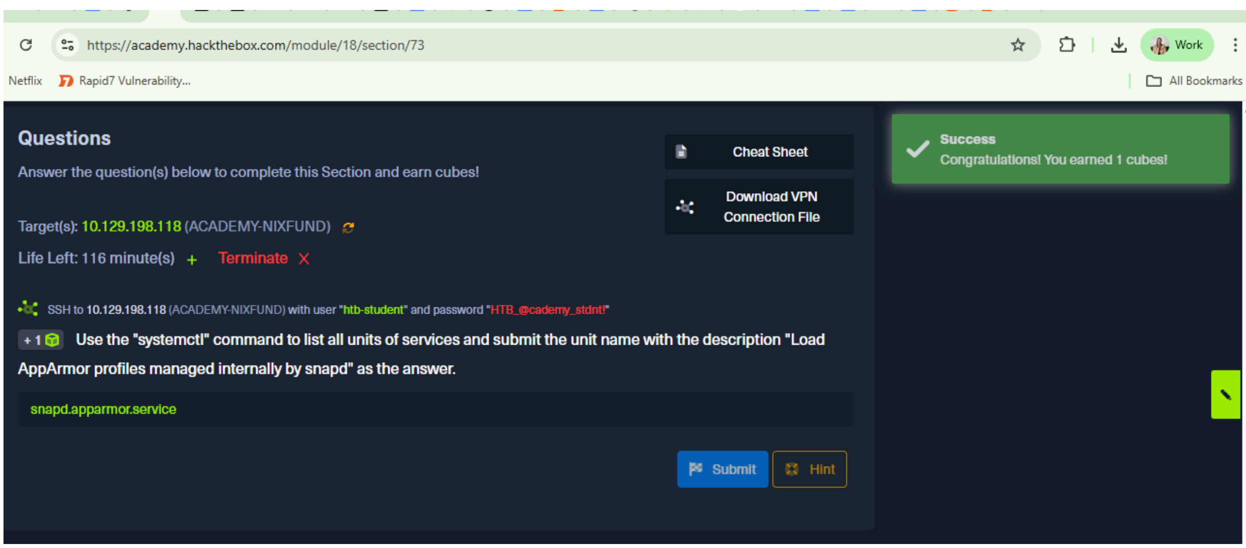
Question

Use the "systemctl" command to list all units of services and submit the unit name with the description "Load AppArmor profiles managed internally by snapd" as the answer.

Answer

Snapd.apparmor.service



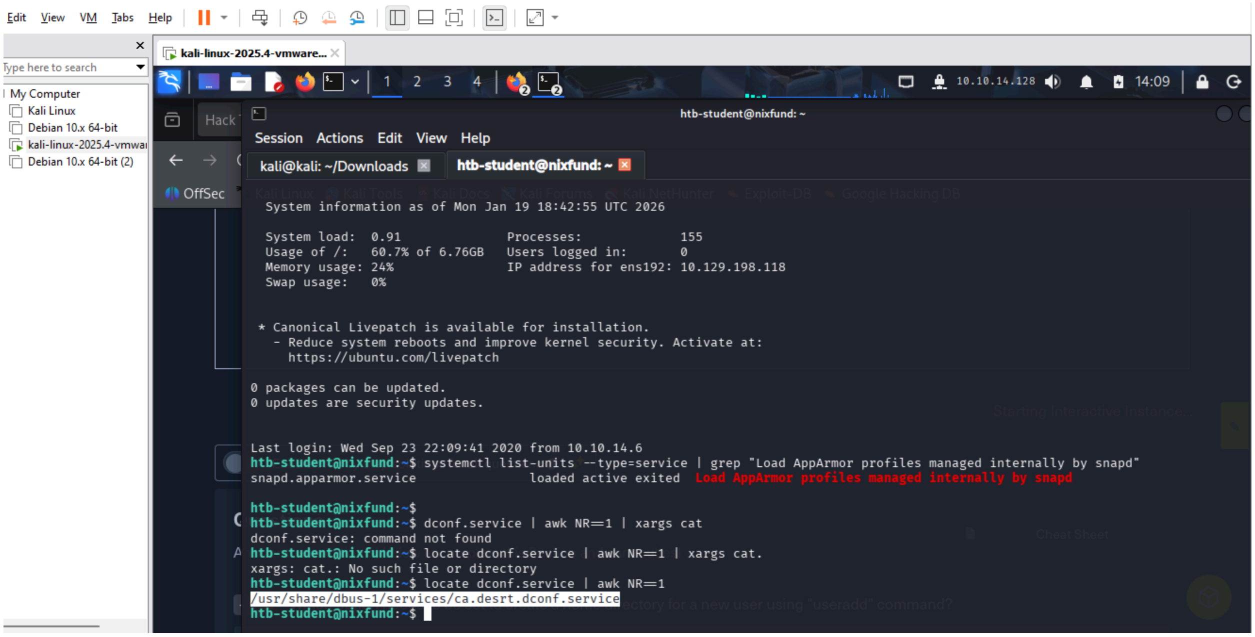


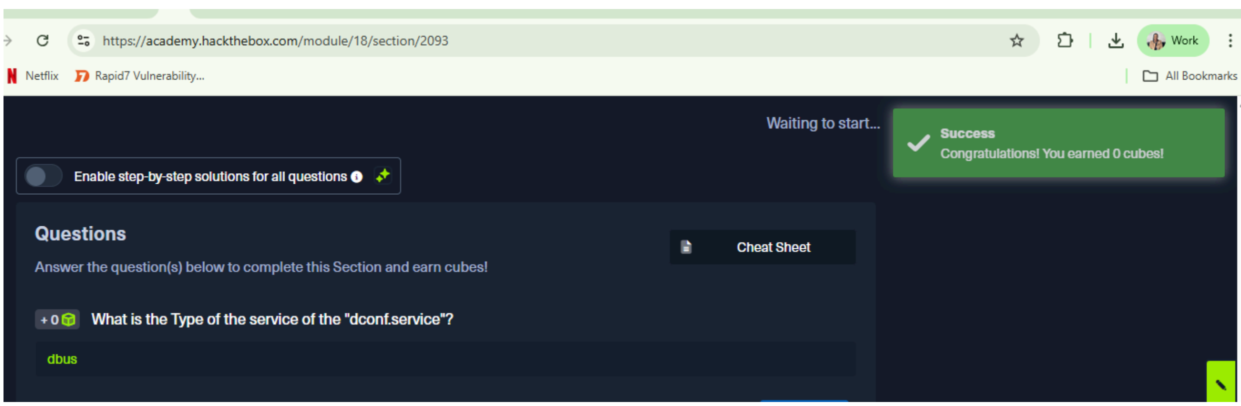
2.7 Task Scheduling

Linux automates tasks using **Systemd** (timers + services) or **Cron** (crontab). Systemd runs scripts at events or intervals; Cron runs scripts at set times. Both handle updates, backups, or maintenance automatically.

Question
What is the Type of the service of the "dconf.service"?
Answer
dbus

explanation:This service is defined as a **D-Bus activated service**, meaning it starts on demand via D-Bus rather than running continuously





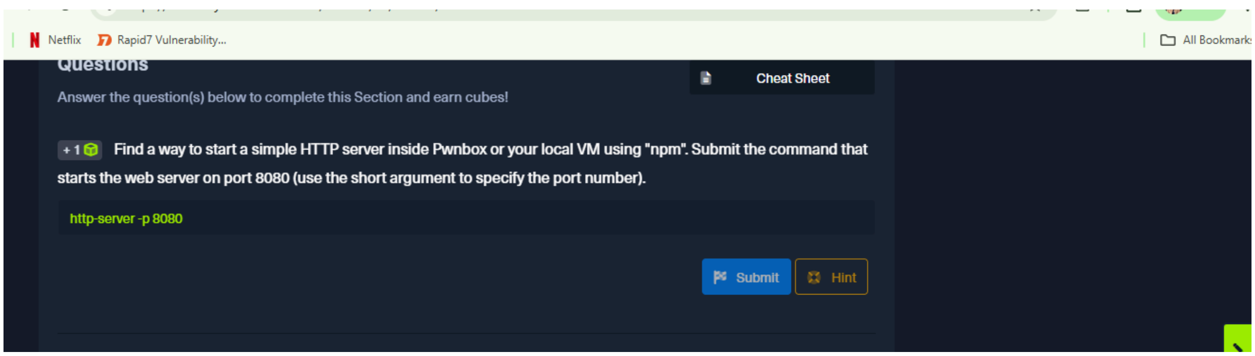
2.8 Network Services

SSH, NFS, Web Server, and VPN are Linux network services used for secure remote access, file sharing, web hosting, and encrypted network connections; install/configure them via apt or Python (openssh-server, nfs-kernel-server, apache2, python3 -m http.server, openvpn) and manage settings in their respective config files for administration and pentesting.

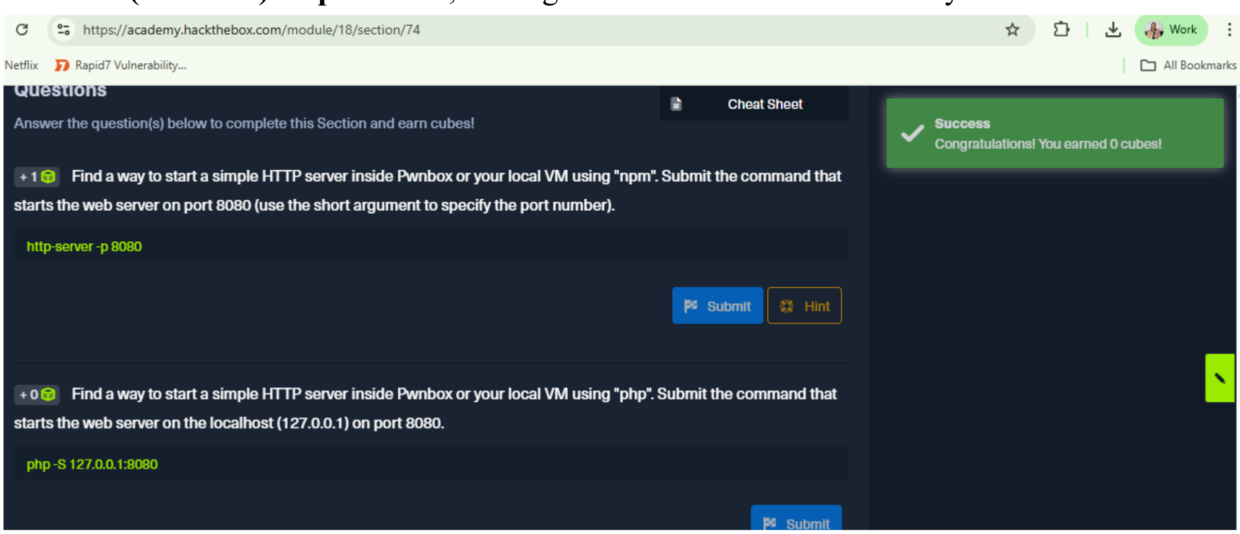
2.9 Working with Web Servers

Web services let browsers and servers communicate; Apache hosts sites, supports modules (e.g., **mod_ssl**, **mod_proxy**) and dynamic content, runs on port 80 by default, and can be tested with **curl**, **wget**, or python3 -m http.server.

Questions
Find a way to start a simple HTTP server inside Pwnbox or your local VM using "npm". Submit the command that starts the web server on port 8080 (use the short argument to specify the port number).
Answer:http-server -p 8080
Explanation:http-server -p 8080 starts a simple Node.js HTTP server and uses the **short -p option** to listen on **port 8080**, serving files from the **current directory**.



Question
Find a way to start a simple HTTP server inside Pwnbox or your local VM using "php". Submit the command that starts the web server on the localhost (127.0.0.1) on port 8080.
Answer:php -S 127.0.0.1:8080
Explanation:
This uses PHP's built-in development server (-S) to start a simple HTTP server bound to **localhost (127.0.0.1)** on **port 8080**, serving files from the current directory



3.0 Back up and Restore

Linux backup and restore can be done efficiently using tools like rsync (fast), Duplicity (encrypted), or Deja Dup (GUI). To back up a directory, use rsync -avz -e ssh /source/

user@backup:/backup/ and to restore, rsync -av user@backup:/backup/ /source/. Backups can be automated with cron jobs.

3.1 File System Management

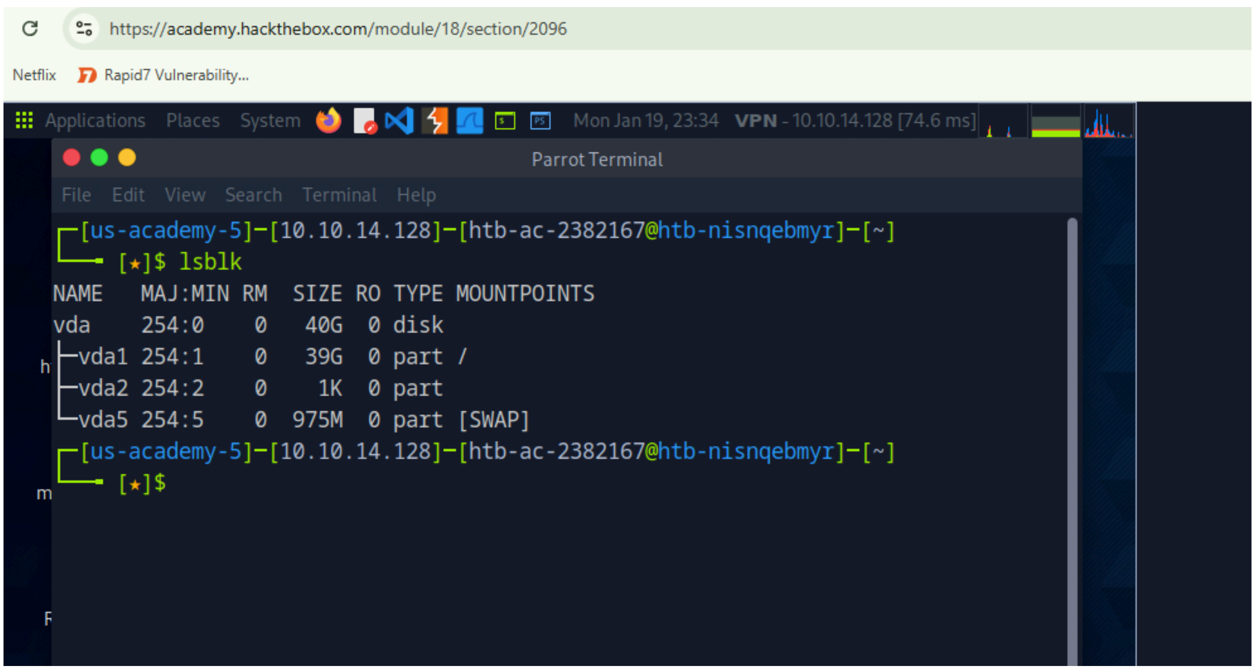
Linux file system management involves choosing appropriate file systems, organizing data using a hierarchical structure with inodes, managing disks and partitions, mounting/unmounting storage, and handling swap space to ensure performance, reliability, and efficient storage use.

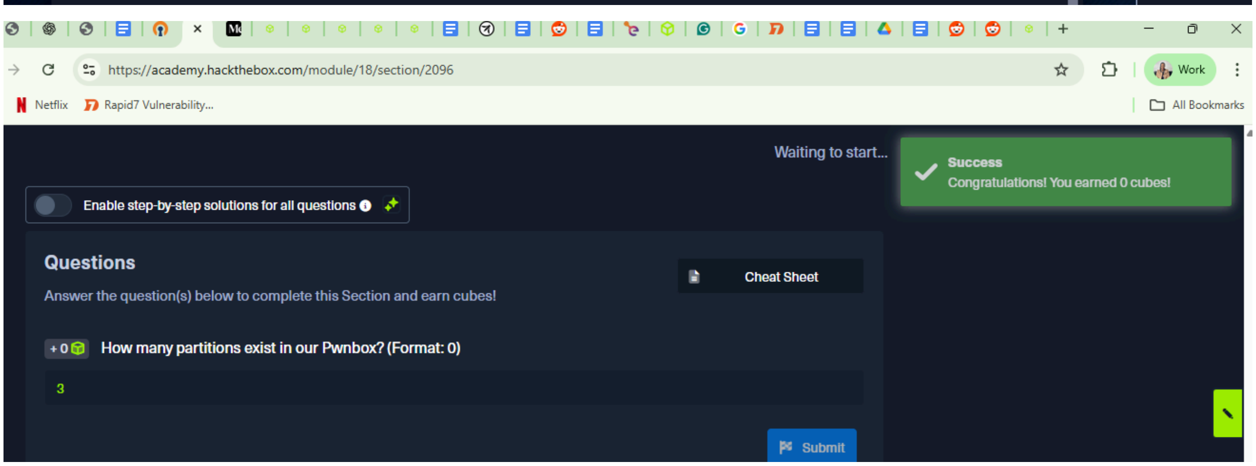
Question

How many partitions exist in our Pwnbox? (Format: 0) :

Answer: 3

Explanation: I spawned and got the 3 partitions, vda1,vda2,vda5.





3.2 Containerization
Containerization packages applications with their dependencies into isolated, lightweight environments like Docker or LXC, enabling consistent, portable, and scalable deployment while requiring proper security measures.

3.3 Network Configuration.
As a penetration tester, mastering Linux network configuration, access control (SELinux, AppArmor, TCP wrappers), and monitoring tools is essential for securing systems, troubleshooting issues, and performing effective testing.

3.4 Remote desktop protocols in Linux
VNC and X11 let Linux admins access remote desktops; VNC is encrypted, X11 needs SSH tunneling for security. Administrators configure servers, set passwords, and manage display ports, ensuring safe and efficient remote management of Linux systems.

3.5 Linux Security
Linux security relies on updates, firewalls, least-privilege access, and tools like SELinux, AppArmor, and fail2ban. Unnecessary services should be removed, strong passwords enforced, and TCP wrappers used to control service access. Security is an ongoing process requiring careful configuration and monitoring.

3.6 Firewall Set up
Linux firewalls (iptables, nftables, UFW) control network traffic using tables, chains, rules, matches, and targets to allow, block, or log packets. They can filter by IP, port, or protocol, create custom chains, and manage traffic, providing flexible security for Linux systems.

3.7 System Logs
Linux system logs track activity for monitoring, troubleshooting, and security. Key logs include: **kernel** (/var/log/kern.log), **system** (/var/log/syslog), **auth** (/var/log/auth.log), **application** (e.g., Apache/MySQL), and **security** (Fail2ban/UFW). Regular review helps detect attacks, misconfigurations, and suspicious activity.

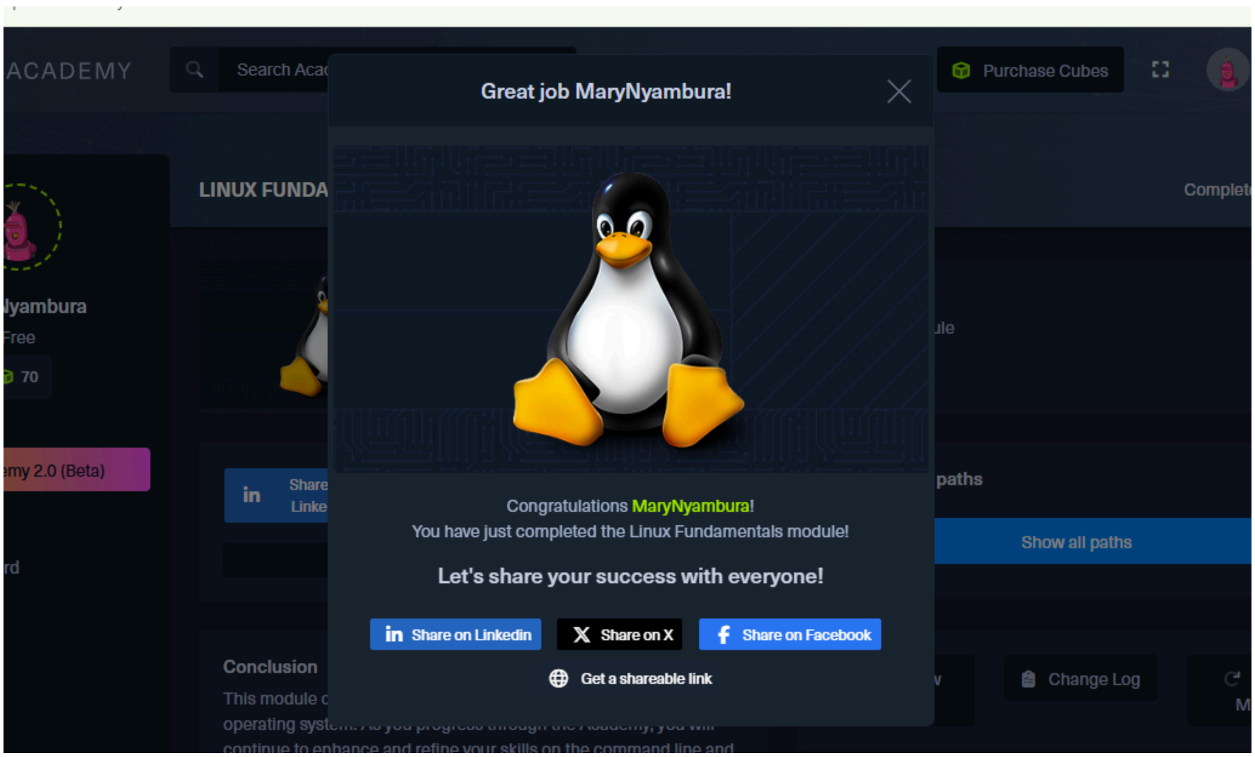
3.8 Solaris
Solaris is a stable, scalable Unix-based OS used in enterprise environments. It features RBAC, SMF, and IPS package management, with proprietary commands for system info, packages, and

process tracing. Optimized for high-end hardware, virtualization, and secure, mission-critical applications.

Link for my completion of the module

https://academy.hackthebox.com/achievement/2382167/18

Screenshot displaying my completion of the module.



Conclusion

In learning Linux fundamentals, I've realized how important it is to understand how the system works from the ground up. From managing file systems and swap space, handling users and permissions, to backing up data securely with tools like rsync or automating tasks with cron, everything connects to keeping a system reliable and safe. I also see the power of containers with Docker and LXC, letting me create isolated environments to test or deploy applications without messing up my main system. Overall, knowing these basics makes me more confident in managing Linux, whether it's for personal projects or real-world environments.